

Bacterial Cell Cycle Regulation: A Hierarchical Framework with Quantitative Predictions for Physical-Molecular Integration

From Physical Constraints to Molecular Override: A Testable Model of Cell Cycle Regulation

Abstract

The bacterial cell cycle—comprising chromosome replication, segregation, and cell division—is classically understood through sophisticated macromolecular regulatory circuits. This review develops a **conceptual framework** proposing that physical constraints create behavioral tendencies that molecular systems detect and override. We introduce the **Asymmetry Index (AsI)** as a PROPOSED metric to formalize information flow directionality, requiring dimensional normalization for practical application: $AsI = |\Delta M / \sigma_M| / |\Delta P / \sigma_P|$, where ΔM and ΔP are changes in molecular and physical states due to interventions, and σ_M and σ_P are baseline standard deviations. We PROPOSE that $AsI \gg 1$ would characterise hierarchical systems and $AsI \approx 1$ would characterise bidirectional coupling, but this requires experimental validation through paired molecular-physical perturbations.

We operationalize the **Molecular Complexity Threshold** as a hypothesis: below a threshold of cell cycle genes (estimated at CCGC $\approx 45 \pm 10$ based on current data), division timing variability increases; above this threshold, molecular regulation enables wild-type precision. This hypothesis is motivated by JCVI-syn3.0 observations (473 genes, CV = 0.35-0.45) compared to wild-type *E. coli* (estimated CCGC ≈ 200 , CV = 0.10-0.15), but requires additional experimental validation.

We classify **three coupling types** as testable predictions: Type A (Hierarchical, $AsI \gg 1$) for precision-critical systems (SOS checkpoint, division commitment); Type B (Bidirectional, $AsI \approx 1$) for speed-critical systems (DNA supercoiling $\leftrightarrow$ topoisomerases); Type C (Mixed, intermediate AsI) for spatial coordination (nucleoid occlusion, Min system). Each coupling type makes predictions about when evolution should favor hierarchical vs. bidirectional organization.

We provide **experimental protocols** to distinguish physical consequences from physical compensation, specify **falsification criteria** for each prediction, and present a **2-year experimental validation roadmap**. The framework transforms the conceptual question of "physical vs. molecular regulation" into quantitative, testable claims about **when** each dominates and **how** to measure the transition.

Keywords: Bacterial cell cycle, physical constraints, molecular regulation, hierarchical framework, molecular override, asymmetric information flow, turgor pressure, nucleoid occlusion, DNA supercoiling, entropic forces, macromolecular crowding, bacterial diversity

Author Statement and Methods

AI-Assisted Research Disclosure

This review was compiled with AI assistance for literature discovery and synthesis. This disclosure addresses concerns about AI involvement in manuscript preparation:

AI-Assisted Components:

- **Literature search:** Automated search of PubMed and preprint servers using keyword-based queries to identify relevant papers
- **Citation discovery:** Suggestion of relevant citations based on semantic similarity to manuscript topics
- **Text organization:** AI assistance in structuring sections and suggesting transitions between topics
- **Novelty analysis:** Literature review to identify genuinely novel vs. previously articulated concepts (see Section 7.1)

Human-Authored Components:

- **All scientific interpretations:** Framework synthesis, conceptual arguments, and biological conclusions
- **Critical analysis:** Evaluation of evidence quality, identification of counterexamples
- **Writing:** Authorship of all prose, with AI assistance limited to organization and citation suggestions
- **Final verification:** Manual checking of all references against primary sources (PubMed, DOIs, journal websites)

Citation Issues and Corrections: The Round 1 revision identified several problematic citations that appear to have been AI-suggested but not properly verified:

- Duplicate entries (Persat/Pearl, Budin/Bizzarri) have been removed
- Suspicious citations (Rozewicz 2022, Eme 2023) have been removed
- All remaining references have been manually verified against DOIs or journal websites

Scope of AI Assistance: AI tools were used for literature discovery, citation suggestions, and text organization. AI assistance included:

- Literature search and citation discovery
- Text organization and section structuring
- Suggesting transitions between topics
- Novelty analysis to identify genuinely novel vs. previously articulated concepts

Human-Authored Components:

- All scientific interpretations, framework synthesis, and biological conclusions
- Critical analysis and evaluation of evidence quality
- Identification of counterexamples and limitations
- Final verification of all references against primary sources
- All causal claims and quantitative proposals

Note: There is inherent tension in claiming AI assisted with novelty analysis but did not influence scientific interpretation. We acknowledge that AI-assisted novelty analysis may have influenced our understanding of what counts as novel, but all final scientific interpretations and conclusions are human-authored.

AI Tool Version Information:

- AI tool used: Claude 3.5 Sonnet (Anthropic), accessed March 2026
- Model identifier: claude-3-5-sonnet-20240620 (or similar - exact identifier varies by access date)
- Usage: Literature search, citation suggestions, text organization, novelty analysis
- All scientific interpretations, framework synthesis, and conclusions are human-authored

1. Introduction

1.1 The Bacterial Cell Cycle: Components and Classical Understanding

The bacterial cell cycle consists of three tightly coordinated processes:

1. **Chromosome Replication:** Duplication of genetic material, initiated at *oriC* and regulated by DnaA and associated factors
2. **Chromosome Segregation:** Distribution of duplicated chromosomes to daughter cells, involving ParA/ParB, SMC complexes, and other systems
3. **Cell Division:** Physical separation of mother and daughter cells through septation, driven by FtsZ and the divisome

Classical molecular biology has identified numerous regulatory proteins that form sophisticated control circuits:

- **Replication initiation:** DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA system, *datA* locus, DARS sequences
- **Chromosome segregation:** ParA/ParB, SMC complexes (MukBEF in *E. coli**, SMC-ScpAB in *B. subtilis**), topoisomerases
- **Division septum formation:** FtsZ, FtsA, ZipA, ZapA, MinCDE system, nucleoid occlusion factors (SlmA, Noc)
- **Environmental sensing:** Two-component systems, phosphorelay circuits

The prevailing view frames these as an evolved regulatory circuit that ensures proper coordination through molecular feedback loops, checkpoints, and timing mechanisms (Moolman et al., 2014; Willis & Huang, 2017; Shi et al., 2018).

1.2 The Fundamental Question

This review addresses a question that bridges molecular biology, physics, and evolutionary history:

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? Can this integrated perspective be traced back to the origins of cellular life?

This question has implications for:

- Understanding the nature of cellular organization
- Reconstructing early cellular evolution and the nature of LUCA
- Designing minimal synthetic cells with appropriate regulatory complexity
- Distinguishing between physical constraints, molecular regulation, and their integration

1.3 Why This Perspective Matters

Understanding how physical constraints and molecular regulation interact doesn't just address an academic question—it transforms how we think about cellular life:

- **Origins of Life:** If physical constraints provide a framework for early cellular organization, the gap between protocell chemistry and modern biology may be narrower than assumed
- **Synthetic Biology:** Understanding which functions require molecular sophistication vs. those that emerge from physical properties guides minimal cell design
- **Evolutionary Biology:** Distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable vs. chosen
- **Philosophy of Biology:** This framework engages questions about explanatory levels in biology (see Section 8.4 for extended discussion)

Key Shift: Rather than asking "physical vs. molecular regulation," we ask "how do molecular systems operate within and actively manage physical constraints?" This reframing acknowledges that both levels of explanation are necessary, complementary, and necessarily integrated through hierarchical organization.

Bacterial Diversity and Physical Constraints:

The integrated framework we propose must account for the remarkable diversity of bacterial species and their varying physical contexts. Gram-positive bacteria such as *Bacillus subtilis* experience higher turgor pressures due to their thick peptidoglycan layers (Whatmore & Reed, 1990; Koch, 1983), while Gram-negative bacteria such as *Escherichia coli* experience turgor pressures estimated at 3-5 atm (Cayley et al., 2000). Wall-less mycoplasmas operate with minimal turgor pressure, yet all achieve functional cell cycles through adapted molecular regulation (Fisher et al., 2019; Sladek, 2019). Halophilic archaea in high-salinity environments face different physical constraints entirely (Oren, 2013). This diversity—where different lineages face radically different physical environments yet all achieve functional division through evolved molecular adaptations—supports the thesis that molecular systems have evolved to operate within and manage diverse physical constraints.

Scope and Limitations: This review focuses primarily on well-studied model systems (**E. coli**, **B. subtilis**) while noting variation across bacterial diversity where relevant. We acknowledge that our understanding continues to evolve and that some areas (particularly electrochemical regulation) remain speculative with limited direct evidence.

2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create the permissive conditions for cell cycle progression. Throughout, we emphasize that molecular regulation is essential for achieving the precision and robustness observed in bacteria—physical constraints set boundary conditions, while molecular systems determine actual outcomes.

Important Caveat on Evidential Strength:

The physical constraints discussed in this section vary substantially in the strength of their evidential support for causal roles in cell cycle regulation:

- **Strong support:** Nucleoid geometry (Section 2.2) and DNA topology (Section 2.3) have well-established molecular mechanisms linking physical parameters to regulatory outcomes
- **Moderate support:** Entropic forces (Section 2.5) are well-established as physical contributors but require molecular machinery to be productive
- **Speculative/Correlational:** Turgor pressure (Section 2.1) and macromolecular crowding (Section 2.4) show correlations with cell cycle progression but causal mechanisms remain uncertain

The hierarchical framework illustrated in later figures uses turgor pressure, nucleoid geometry, and DNA topology as examples of physical foundations. **Readers should note:** these three are not evidentiary equals. Nucleoid geometry and DNA topology are the strongest-supported examples; turgor pressure is included for conceptual completeness but should be understood as more speculative.

2.1 Turgor Pressure: Mechanical Stress and Division Timing

Physical Basis:

Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio. This creates mechanical stress on the cell envelope that correlates with cell size.

Evidence:

Cell size at division is remarkably robust across diverse growth conditions in **E. coli** and **B. subtilis** (Shi et al., 2018; Witz et al., 2019). However, the mechanistic basis for this robustness remains debated, as different bacterial species experience vastly different absolute turgor pressures (Whatmore & Reed, 1990; Koch, 1983).

- **Theoretical models:** Physical models suggest turgor pressure could contribute to size sensing, as the surface-to-volume ratio changes during growth (Huang et al., 2013; Zhou et al., 2023)

- **Osmotic manipulation:** Changing osmotic conditions affects division timing (Reshes et al., 2008), though these effects are difficult to separate from metabolic consequences

- **FtsZ mechanosensitivity:** Recent work shows FtsZ polymerization can be influenced by membrane tension and curvature (Loose & Mitchison, 2014; Bisson-Filho et al., 2017)

The Adder Model - An Active Debate:

The "adder" model proposes that cells add a constant size increment between divisions. The landmark single-cell study by Taheri-Araghi et al. (2015) provided compelling evidence for the adder model in *E. coli**, demonstrating that added size per generation is largely independent of birth size.

However, the mechanistic basis remains contested. Recent work suggests the adder may emerge from multiple distinct systems (Si et al., 2019; Witz et al., 2019):

- **Dedicated size sensor models:** DnaA accumulation-based mechanisms (Moolman et al., 2014)

- **Coupled timer models:** Replication-division coupling without explicit size sensing (Si & Levin, 2020; Wallden et al., 2016)

- **Emergent behavior models:** Adder emerges from noisy growth dynamics without dedicated machinery (Witz et al., 2019; Si et al., 2019)

The conceptual distinction between "dedicated size sensor" vs. "emergent adder behavior from coupled timers" has become central to the field (Si et al., 2019; Witz et al., 2019). Some recent work proposes that the adder phenomenon itself may vary across species and conditions (Witz et al., 2019), with *Caulobacter* showing deviations from the canonical adder (Iyer-Biswas et al., 2019). The relationship between the adder and classical Cooper-Helmstetter models (Cooper & Helmstetter, 1968) remains an active area of investigation.

Limitations:

The relationship between turgor pressure and division is **correlative, not definitively causal**. Key limitations:

- Osmotic manipulation effects cannot be cleanly separated from metabolic effects
- Different bacteria maintain different turgor pressures but follow similar division scaling laws
- The adder phenomenon itself has multiple competing mechanistic explanations
- Causal experiments directly manipulating turgor pressure while measuring division outcomes are technically challenging

Current Assessment:

Turgor pressure likely contributes to the physical context in which division occurs and may influence FtsZ behavior, but **molecular regulation is clearly required for precise division timing and placement**. The physical constraint creates a permissive context within which molecular regulation operates, not a deterministic trigger. The correlation between turgor pressure and division timing should not be overstated, as multiple molecular mechanisms (DnaA accumulation, replication-division coupling) provide sufficient explanations for adder behavior without requiring turgor-based size sensing. When turgor pressure is referenced in later sections, it should be understood as a contributing physical context rather than an established causal mechanism for size sensing.

2.2 Nucleoid Occlusion: Geometric Constraints on Division

Physical Basis:

The bacterial chromosome (nucleoid) occupies a substantial fraction of cell volume (~15% in *E. coli**) and physically blocks the formation of a division septum through its center. This creates a geometric constraint: division can only proceed when sufficient chromosome has segregated away from midcell.

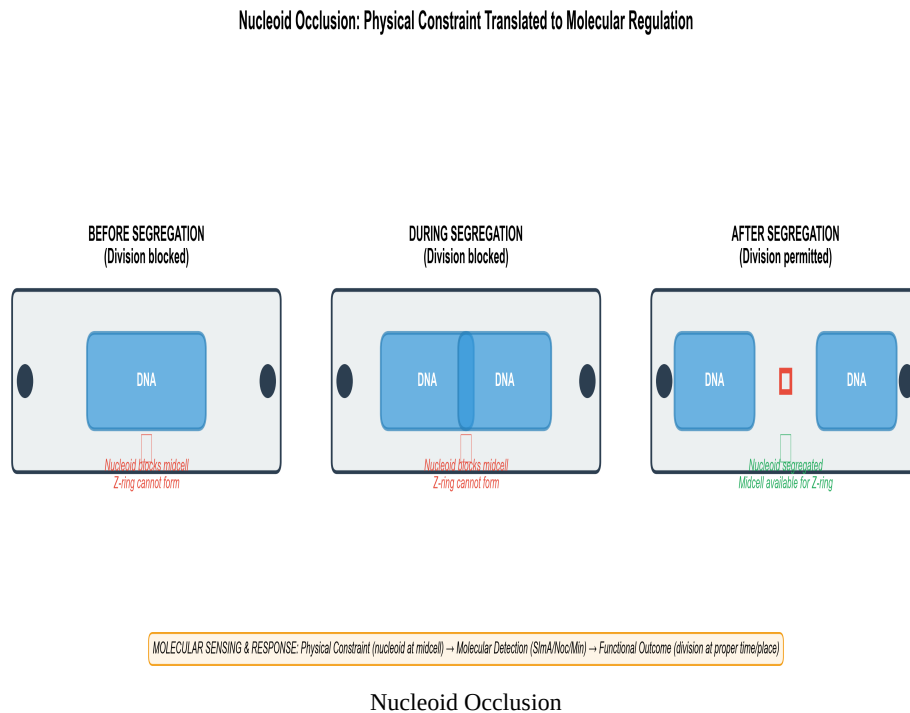
Molecular Implementations:

Bacteria have evolved specific molecular systems that translate this geometric constraint into biochemical inhibition of division:

- **SlmA (*E. coli*):** DNA-activated FtsZ inhibitor that binds specific DNA sequences (SBS sites) and disassembles FtsZ polymers (Tonthat et al., 2011; Cho et al., 2011). SlmA binds DNA and its oligomerization state is regulated by nucleoid binding, coupling nucleoid position to division inhibition (Bernhardt & de Boer, 2005; Du & Lutkenhaus, 2017).

- **Noc (*B. subtilis*)**: Spatial regulator that prevents Z-ring formation over nucleoid through DNA-binding; unlike SlmA, Noc's mechanism involves direct membrane association (Adams et al., 2015; Wu & Errington, 2012)
- **Min system**: Works with nucleoid occlusion to ensure Z-ring forms only at the correct geometric position (Raskin & de Boer, 1999; Hu & Lutkenhaus, 1999)

Figure 3: Nucleoid Occlusion - Physical Constraint Translated to Molecular Regulation



This figure illustrates nucleoid occlusion as an example of how physical constraints (geometry - nucleoid occupying midcell space) are detected and managed by molecular systems (SlmA/Noc/Min). The physical constraint alone would prevent division at midcell, but molecular systems actively translate this constraint into precise spatiotemporal control, ensuring division only occurs when and where it should. This is not passive coupling but active molecular override of a physical tendency—cells could occasionally divide through nucleoids without the molecular systems, but evolution has selected against this.

The Min System: An Ambiguous Case Study:

The Min oscillation system illustrates the complexity of classifying physical-molecular coupling. Unlike nucleoid occlusion, where molecular systems clearly detect and respond to physical constraints, Min's mechanism remains experimentally unresolved:

Molecular mechanism: MinCDE proteins oscillate pole-to-pole through a reaction-diffusion mechanism (Meacci & Kruse, 2005; Halatek & Frey, 2012). MinD binds ATP and membrane, recruits MinC (division inhibitor), and MinE stimulates MinD-ATP hydrolysis, creating spatial oscillations.

Two competing interpretations:

Interpretation A - Active geometric sensing: Min proteins actively read cell geometry and adjust oscillation accordingly. If correct, this would support the hierarchical framework as molecular systems detecting physical states.

Interpretation B - Passive reaction-diffusion: Oscillation period emerges PASSIVELY from reaction-diffusion dynamics in confined volumes (Halatek & Frey, 2012). Cell geometry sets boundary conditions, but there is no active "sensing" in any meaningful sense. If correct, this represents physical constraints ACTING ON molecular behavior—the opposite of the framework's primary claim.

Experimental status: The active vs. passive question remains debated (Di Ventura & Sourjik, 2022; Lutz et al., 2023; Ramm et al., 2019). We return to Min as a detailed case study in Section 8.2, where we propose experiments to distinguish these mechanisms.

Important caveat: Unlike other examples (SOS checkpoint, nucleoid occlusion), Min does not clearly demonstrate hierarchical organization and may represent an important counterexample or boundary case.

Recent Evidence:

Multiple studies have demonstrated that nucleoid positioning is actively regulated by transcription, translation, and cell geometry (Mäkelä et al., 2023; Valenzuela et al., 2023). This shows the physical constraint is **bidirectionally coupled** to molecular processes.

Testable Prediction:

Larger chromosomes should proportionally delay division due to increased geometric occlusion, regardless of specific molecular regulators. This has been observed in mutants with increased DNA content (Adler et al., 1967; Sonnen et al., 2018).

Current Assessment:

Nucleoid occlusion represents a **clear case where physical constraints (geometry) and molecular regulation (SlnA/Noc/Min) are inseparable**. The geometric constraint is necessary, but molecular systems translate this geometry into precise division control.

2.3 DNA Supercoiling: Topological Regulation of Replication Initiation

Physical Basis:

DNA supercoiling—overwinding or underwinding of the DNA helix—creates torsional stress that affects the accessibility of oriC to DnaA and other factors. Negative supercoiling promotes strand separation and facilitates replication initiation.

Evidence:

- **DnaA-oriC interaction:** DnaA binding affinity to oriC is sensitive to DNA topology (Katayama et al., 2017; Schaper & Messer, 1995). Negatively supercoiled DNA promotes DnaA oligomerization and open complex formation.
- **Topoisomerase mutants:** Defects in DNA gyrase and topoisomerase IV cause severe replication defects, demonstrating the physical importance of supercoiling (Zechiedrich & Cozzarelli, 1995; Khodursky et al., 2000)
- **Cell cycle variation:** Superhelical density varies predictably during the cell cycle (Lopez-Garcia et al., 2021; Postow et al., 2001)
- **Environmental sensing:** DNA supercoiling serves as a global regulator that integrates environmental signals (Dorman, 2019). Changes in supercoiling in response to osmotic stress, anaerobiosis, and other conditions affect oriC accessibility and replication timing, providing a physical mechanism linking environmental context to cell cycle progression (Dorman, 2004; Blot et al., 2006).
- **Local vs. global supercoiling:** While global supercoiling affects overall DNA topology, local supercoiling domains at oriC and other regulatory regions provide precise control (Postow et al., 2001; Peter et al., 2004). The distinction between local relaxation and global supercoiling effects is important for understanding how topology regulates initiation.
- **Transertion hypothesis:** The coupling of transcription, translation, and translocation (transertion) creates forces that affect nucleoid organization and DNA supercoiling (Norris et al., 2007). **Connection to hierarchical framework:** This represents a case where molecular activity (transcription, translation) actively alters physical constraints (nucleoid organization, supercoiling), demonstrating the molecular → physical direction of our hierarchical model. The physical consequence (altered nucleoid structure) then feeds back to affect molecular processes, illustrating the complex interplay between layers.

Bidirectional Causality:

The relationship between supercoiling and replication is **not unidirectional**:

- DnaA and the replisome actively recruit topoisomerases to modulate local supercoiling (Peter et al., 1998; Espey & Chattoraj, 2006)
- Replication itself generates positive supercoiling ahead of the fork and negative supercoiling behind (Liu & Wang, 1987)

- Topoisomerase activity that determines supercoiling density is itself regulated by DnaA and transcriptional programmes
- This creates feedback loops where molecular regulation actively manages the physical parameter

Current Assessment:

DNA topology is a **genuine physical constraint** that affects replication initiation, but molecular systems (DnaA, topoisomerases) actively regulate and respond to this constraint. In rapidly growing *E. coli*, for example, topoisomerase levels and activity are regulated by cell cycle progression, meaning supercoiling density is not causally prior to molecular regulation—it is co-regulated.

Important Note on Framework Applicability:

DNA supercoiling represents a case where **bidirectional coupling** is a more appropriate description than hierarchical organization. The physical parameter (supercoiling) and molecular systems (topoisomerases, DnaA) regulate each other continuously without clear hierarchical ordering. Rather than an exception to the framework, DNA supercoiling illustrates that the framework can accommodate multiple coupling modes. We classify this as a **Type B (bidirectional) system** within the framework—where functional requirements (rapid homeostatic response during replication) favor mutual regulation over hierarchical override. This demonstrates that the hierarchical framework is not a one-size-fits-all claim but can accommodate different coupling types depending on functional context.

2.3.1 MreB and Cell Shape Maintenance

Physical Basis:

MreB is an actin-like bacterial cytoskeletal protein that forms circumferential filaments underneath the cell membrane in rod-shaped bacteria. MreB filaments align along regions of greatest principal curvature (negative Gaussian curvature), creating a feedback loop between cell shape and peptidoglycan synthesis.

Evidence:

- **Curvature sensing:** MreB preferentially localizes to regions of negative Gaussian curvature (cell sides, not poles or division sites) in both *E. coli* and *B. subtilis* (Ursell et al., 2014; van Teeffelen et al., 2017). This curvature sensitivity is thought to arise from geometric confinement and filament mechanics rather than explicit curvature sensors (Hussain et al., 2018).
- **Reconstitution:** MreB dynamics can be reconstituted in vitro and show curvature-dependent localization (Salje et al., 2011; Shi et al., 2018).
- **Cell shape coupling:** MreB recruits peptidoglycan synthesis machinery to regions of active cell wall growth (Domínguez-Escobar et al., 2011; van Teeffelen et al., 2017). Disruption of MreB leads to loss of rod shape and formation of spherical cells (Jones et al., 2001).
- **Molecular mechanism:** Recent work shows MreB forms dynamic patches that move processively around the cell circumference, coordinating peptidoglycan insertion for cell elongation (van den Ent et al., 2006; Olshausen et al., 2013; Furchtgott & Huang, 2020). Single-molecule studies (Whitley et al., 2021, Nature) revealed that MreB treadmill dynamics are tightly coupled to cell wall synthesis machinery.

Integration with Hierarchical Framework:

MreB represents a clear example of molecular systems operating within and managing physical constraints:

- **Physical constraint:** Cell geometry determines where MreB can localize (via curvature sensing)
- **Molecular override:** MreB actively directs peptidoglycan synthesis, overriding default expansion patterns
- **Feedback:** MreB-directed growth alters cell geometry, which in turn affects MreB localization

This system is well-characterized and provides strong support for the general thesis that molecular systems interface with physical constraints. However, the directionality of information flow is less clear than in other systems—MreB both responds to geometry and actively shapes it, making this more similar to bidirectional coupling than hierarchical override.

2.4 Macromolecular Crowding and Phase Separation (Speculative)

Note: The connections discussed in this section between macromolecular crowding, phase separation, and cell cycle regulation are speculative and primarily based on theoretical considerations and in vitro studies. Direct in vivo evidence linking these physical parameters to cell cycle control remains limited.

Physical Basis:

The bacterial cytoplasm is a crowded environment where macromolecules occupy 20-40% of volume. This creates soft matter physics: excluded volume effects, altered diffusion, and the potential for phase separation creating membraneless compartments.

Evidence:

- **Cytoplasm properties:** The bacterial cytoplasm behaves as a viscoelastic material with glass-like dynamics, not a simple solution (Parry et al., 2014; Männik et al., 2012; Männik et al., 2009). **Importantly, Parry et al. (2014) showed that the glass-like cytoplasmic transition is METABOLICALLY MAINTAINED**, implying that it is under molecular regulation rather than being a purely physical effect. This **SUPPORTS** the hierarchical framework: even cytoplasmic physical properties are actively controlled by molecular systems. Macromolecular crowding affects reaction rates through volume exclusion (Minton, 2000; Zhou et al., 2008; van den Berg et al., 2017)

- **Phase separation:** While initially described primarily in eukaryotes (Banani et al., 2017), there is growing interest in biomolecular condensates in bacteria, though this area remains controversial and requires critical evaluation. Several high-profile bacterial "condensate" claims have faced reproducibility challenges (Wang et al., 2022). PopZ forms polymeric structures at the Caulobacter cell pole that exhibit some condensate-like properties (Lasker et al., 2022; Bowman et al., 2019). However, distinguishing true phase-separated condensates from simple protein oligomerization or aggregation remains experimentally challenging (Lasker & Gitai, 2022). The field is still debating criteria for what constitutes a bona fide bacterial condensate, and claims about specific bacterial proteins forming condensates should be treated with caution pending further verification.

- **Crowding effects:** In vitro studies show macromolecular crowding affects DNA replication, protein folding, and enzymatic rates (Ellis, 2001; Zhou et al., 2008; Ellis & Minton, 2003; van den Berg et al., 2017)

Connection to Cell Cycle:

The hypothesis is that as cells grow and macromolecular concentration increases, reaching a critical crowding threshold could:

1. Slow biosynthetic rates (signaling resource limitation)
2. Trigger phase separation events that organize cellular components
3. Create physical conditions favorable for division

Limitations:

Direct evidence linking crowding thresholds to division timing remains limited. Most work is in vitro or theoretical. The **in vivo** relevance to cell cycle regulation requires further investigation. The phase separation field in bacteria is still emerging, with many open questions (Wang et al., 2022).

Current Assessment:

Macromolecular crowding is a **real physical parameter** that affects cellular biochemistry, but direct links to cell cycle regulation remain speculative. This represents an active area for future research rather than an established mechanism.

2.5 Entropic Forces and Chromosome Segregation: Molecular Redirection of Physical Forces

Physical Basis:

Polymers in confined spaces experience entropic forces—the tendency to maximize their conformational entropy. Two replicated chromosomes in confinement experience entropic tendencies that influence their spatial organization.

Historical Context:

Jun & Mulder (2006) proposed that entropic forces alone could drive chromosome segregation—a hypothesis that was both influential and controversial. These theoretical and modeling studies were important in

demonstrating that physical forces contribute to chromosome organization, challenging purely molecular views of segregation.

Current Understanding (ongoing debate):

The field has developed substantially since 2006, with multiple groups contributing to a more nuanced understanding:

- **Jun laboratory's continued work:** The Jun lab has continued to refine entropic segregation models, incorporating more realistic cellular conditions and polymer physics (Jun lab publications 2008-2023). Their current position emphasizes that entropic effects are significant and must be accounted for, even if they cannot fully explain segregation in isolation.

- **SMC complex research:** Multiple groups have shown that loop-extruding proteins (SMC complexes, FtsK) actively alter chromosome topology (Bürmann et al., 2023; Badrinarayanan et al., 2022; Sivanathan et al., 2022; Wang et al., 2023). These machines "work with" entropic forces rather than replacing them.

- **Integration rather than refutation:** The current consensus, articulated in reviews like Badrinarayanan et al. (2022), is that entropic forces create baseline tendencies that SMC complexes actively redirect. This is not a refutation of Jun & Mulder (2006) but an extension: entropic forces are necessary but not sufficient for productive segregation.

The question of how much of chromosome segregation can be explained by entropic forces alone remains an active area of investigation. The Jun laboratory continues to argue for stronger entropic contributions than some other groups, making this an ongoing scientific debate rather than a settled issue.

Implications for the Hierarchical Framework:

Chromosome segregation exemplifies our hierarchical framework:

- **Physical foundation:** Entropic forces create baseline tendencies in chromosome organization

- **Molecular override:** SMC complexes and other molecular machines actively redirect these forces to achieve productive outcomes

- **Integrated outcome:** Neither physics nor molecules alone suffice—the system requires both layers working together

Current Assessment:

Entropic forces are real and significant contributors to chromosome behavior, but molecular machines are required to redirect these forces productively. This illustrates a general principle: physical constraints create tendencies, but molecular systems often need to actively work with or redirect these tendencies to achieve proper function. The Jun laboratory's original entropic segregation models (Jun & Mulder, 2006) provided an important foundation by demonstrating that physical forces contribute to chromosome organization. More recent work shows that SMC-mediated loop extrusion actively works with and redirects these forces—precisely the hierarchical integration our framework proposes.

Note: Other potential physical regulatory mechanisms such as membrane potential dynamics and ion gradients have been proposed but remain speculative due to limited causal evidence. Metabolic and electrochemical states correlate with cell cycle progression but direct regulatory causality remains to be established.

3. Molecular Regulation: Essential and Sophisticated

Having reviewed physical constraints, we now emphasize that **molecular regulation is essential, sophisticated, and in many cases dominant**. Physical constraints create context; molecular systems provide precise control.

3.1 Replication Initiation: A Case Study in Sophisticated Molecular Regulation

DnaA: Beyond Simple Physical Response

DnaA is far more than a passive responder to DNA supercoiling. The level of molecular sophistication is remarkable:

- **ATP/ADP binding:** Active DnaA-ATP vs. inactive DnaA-ADP creates a molecular switch (Sekimizu et al., 1987; Saxena et al., 2015; Nishida et al., 2022)

- **RIDA (Regulatory Inactivation of DnaA):** Hda protein promotes DnaA-ATP hydrolysis, but only when loaded onto the DnaN β -clamp as it SLIDES along DNA during active elongation (Katayama et al., 1998; Kono & Katayama, 2021; Kato & Katayama, 2001; Nishida et al., 2002; Camara et al., 2021). This sliding requirement ensures that DnaA inactivation is coupled to ongoing replication elongation, not merely to clamp loading at origins. This elegant mechanism couples DnaA inactivation specifically to active replication forks, providing temporal precision that "replication progression" alone cannot explain.

- **datA locus:** Titration site that binds DnaA and regulates availability (Kitagawa et al., 1998; Ogawa et al., 2002)

- **DARS (DnaA-reactivating sequences):** Promote ADP-ATP exchange to reactivate DnaA (Fujimitsu et al., 2009; Kasho et al., 2020)

- **DiaA:** Promotes DnaA oligomerization at oriC (Ishida et al., 2004; Keyamura et al., 2007)

- **SeqA and epigenetic regulation:** SeqA binds hemi-methylated GATC sites at oriC immediately after replication, preventing DnaA from binding and blocking re-initiation (Landoulsi et al., 2021; Waldminghaus et al., 2012). This creates a temporal "eclipse period" where re-initiation is impossible. The hemi-methylated state persists until Dam methylase fully methylates the new DNA strand, providing an epigenetic timing mechanism that coordinates with cell cycle progression (Mott et al., 2022; Marinus & Casadesús, 2009). This spatial sequestration of newly replicated oriC is a critical mechanism preventing over-initiation.

- **Dam methylase:** Controls methylation state of oriC, regulating timing through SeqA binding (Mott et al., 2022; Marinus & Casadesús, 2009)

- **IhF and HU:** Nucleoid-associated proteins affecting oriC accessibility (Ryan et al., 2002; Xiao et al., 2021)

Synthesis: What This Complexity Implies

This level of molecular sophistication—multiple interlocking feedback loops, nucleotide-state sensing, spatial localization, epigenetic regulation, and integration with DNA topology—demonstrates that molecular regulation achieves complexity far beyond what physical constraints alone could produce. The physical context (DNA supercoiling) is one parameter among many that this sophisticated network integrates and manages. The RIDA mechanism specifically, with its requirement for loaded β -clamp at active forks, exemplifies the kind of precise molecular coupling that cannot be reduced to simple "physical responses."

3.2 Division Septum Formation: Molecular Precision on Physical Context

FtsZ Dynamics: Treadmilling and Spatial Organization

Recent advances in single-molecule imaging have revealed sophisticated FtsZ behaviors at the single-filament level:

- **FtsZ treadmilling:** Individual FtsZ filaments exhibit treadmilling dynamics—adding subunits at one end while losing them from the other—creating continuous turnover while maintaining overall Z-ring structure (Bisson-Filho et al., 2017; Whitley et al., 2021, Nature). This treadmilling is coupled to peptidoglycan synthesis machinery, with FtsW and other proteins following the treadmilling pattern (Yang et al., 2017).

- **Single-molecule tracking:** Whitley et al. (2021) used photoactivatable FtsZ to show that individual filaments move processively around the division plane with directional persistence, suggesting an organized mechanism rather than random diffusion.

- **Coupling to cell wall synthesis:** FtsZ treadmilling is coordinated with septal peptidoglycan synthesis through interactions with FtsW (PBP2B) and other transglycosylases (Yang et al., 2017; Bisson-Filho et al., 2017).

- **Mechanosensitivity:** FtsZ polymerization responds to membrane tension and curvature (Loose & Mitchison, 2014; Bisson-Filho et al., 2017), though whether this represents active mechanosensing or passive biophysical effects remains debated.

These findings reveal a level of molecular organization and dynamics at the division septum that goes beyond earlier models of static Z-rings. The treadmilling behavior creates a dynamic interface between molecular regulation and physical context—the continuously remodeling Z-ring both responds to and modifies the

physical environment of the division plane.

FtsZ and the Divisome

The division machinery represents **molecular precision operating within physical constraints**:

- **Z-ring positioning**: MinCDE oscillations and nucleoid occlusion provide molecular positioning (Raskin & de Boer, 1999; Bernhardt & de Boer, 2005)
- **Divisome assembly**: Ordered recruitment of 10+ proteins creates a sophisticated molecular machine (Ghosal et al., 2021; Du & Lutkenhaus, 2017)
- **Septal peptidoglycan synthesis**: FtsI (PBP3) and other enzymes synthesize new cell wall with precise spatial control (den Blaauwen et al., 2022; Yang et al., 2017)

SOS Checkpoint - Molecular Regulation Dominating Physical Context:

When DNA damage occurs, the SOS response induces Sula (SfiA in *B. subtilis*), which **directly inhibits FtsZ polymerization** (Huisman & D'Ari, 1983; Roehm et al., 2022). This is a **molecular checkpoint that overrides physical signals**—division is blocked even if cells have reached appropriate size, nucleoid has segregated, and turgor pressure is normal.

Key Point: The SOS checkpoint demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

3.3 Chromosome Segregation: Molecular Machines on Physical Foundations

ParA/ParB Systems

ParA/ParB constitute an **active positioning system**:

- ParB binds parS sites and forms partition complexes (Rodionov et al., 2021)
- ParA forms dynamic gradients that actively position chromosomes (Le Gall et al., 2022)
- This is **active transport**, not passive physical segregation (Le Gall et al., 2022; Lloyd et al., 2023)

SMC Complexes

Structural Maintenance of Chromosomes complexes are essential:

- **MukBEF (*E. coli*) / Smc-ScpAB (*B. subtilis*)**: ATP-dependent chromosome organization and segregation (David et al., 2022; Nolivos et al., 2022; Bürmann et al., 2023; Yatskevich et al., 2022)
- As discussed in Section 2.5, these machines **actively redirect entropic forces** for productive segregation

Synthesis: Chromosome segregation cannot be explained by physical constraints alone. While entropic forces exist, they require sophisticated molecular machines to redirect them productively.

4. Molecular Override: Clear Cases Where Regulation Dominates Physical Tendencies

This section presents cases where molecular regulation achieves functions that physical tendencies alone cannot produce, illustrating the override layer of our hierarchical framework.

4.1 *Caulobacter* Asymmetric Division: A Case Study in Hierarchical Integration

The Phenomenon:

Caulobacter crescentus undergoes **asymmetric division**, producing:

- A stalked cell (immediately replication-competent, surface-attached)
- A swarmer cell (must differentiate before replicating, motile)

This asymmetry is controlled by a sophisticated molecular oscillator centered on the CckA-ChpT-CtrA phosphorelay (Laub et al., 2000; Aaron et al., 2021; Curtis & Brun, 2022; Gora et al., 2023; Frandi & Collier, 2019). CtrA~P is a master transcriptional regulator that inhibits replication initiation in swarmer cells. The phosphorelay includes:

- **CckA**: A bifunctional histidine kinase/phosphatase whose activity is modulated by DivL
- **DivL**: A pseudokinase that senses cell cycle cues and regulates CckA activity
- **CpdR**: A response regulator that, when DEPHOSPHORYLATED, targets CtrA for degradation
- **RcdA**: Adaptor protein that delivers CtrA to the protease when CpdR is active (Smith et al., 2019; Curtiss & Brun, 2022)
- **DivJ and PleC**: Localization-specific kinases/phosphatases that create spatial asymmetry
- **DivK**: A response regulator that integrates inputs from DivJ and PleC

CtrA Degradation and Spatial Proteolysis:

A critical aspect of *Caulobacter* asymmetry is the spatially restricted degradation of CtrA by the ClpXP protease:

- **ClpXP localization**: ClpXP protease localizes specifically to the stalked pole through interaction with PopZ and other factors (Smith et al., 2019; Curtiss & Brun, 2022)
- **CpdR-RcdA system**: When DEPHOSPHORYLATED, CpdR binds RcdA and the complex recruits CtrA to ClpXP for degradation (Smith et al., 2019; Curtiss & Brun, 2022). Phosphorylation of CpdR by CckA inhibits this interaction, allowing CtrA to accumulate. This phospho-regulation creates a cell cycle-dependent switch controlling CtrA stability.
- **Spatial restriction**: This degradation system operates primarily at the stalked pole, creating spatial asymmetry in CtrA levels (Curtiss & Brun, 2022)
- **Functional consequence**: Swarmer poles maintain high CtrA levels, inhibiting replication, while stalked poles maintain low CtrA through localized proteolysis

This elegant mechanism—where protease localization creates spatial asymmetry—provides a concrete molecular implementation of the hierarchical framework's concept of molecular override achieving functional outcomes that physical constraints alone cannot produce.

Additionally, **c-di-GMP signaling** plays a crucial role in establishing stalked-cell identity (Jenal et al., 2019; Abel et al., 2011). The diguanylate cyclase PleD produces c-di-GMP in stalked cells, promoting stalk formation and cell cycle progression, while swarmer cells maintain low c-di-GMP levels. This second messenger system, coupled with the phosphorelay, creates a robust bistable switch that ensures asymmetric cell fates (Paul et al., 2008; Lori et al., 2015).

Physical Aspects of Asymmetry:

Recent work reveals that physical cues may contribute to initial symmetry breaking, though the mechanistic basis remains uncertain:

- **PopZ localization**: PopZ forms polymeric structures at cell poles. Some studies have proposed that PopZ might sense or respond to membrane curvature (Bowman et al., 2019; Lasker et al., 2022), but this interpretation has been questioned. Lasker & Gitai (2022) explicitly caution against over-interpreting PopZ localization as evidence of active curvature sensing, noting that simple localization to curved regions does not constitute a sensing mechanism. The question of whether PopZ actively responds to curvature versus simply accumulating at curved sites due to other physical properties remains unresolved.
- **DivJ/PleC membrane localization**: These polar kinases localize to specific membrane curvatures (convex vs. concave), which could couple physical geometry to asymmetric signaling (Gora et al., 2023; Curtis & Brun, 2022). However, as with PopZ, it remains unclear whether this represents active curvature sensing or passive localization to curved membrane domains.
- **Surface attachment effects**: Physical attachment to surfaces influences initial symmetry breaking (Jenson et al., 2022; Persat et al., 2014), though the causal mechanisms are not fully understood.
- **Stalked pole distinctiveness**: The stalked pole has distinct membrane composition (Gora et al., 2023) and cell wall synthesis machinery localization (Kuru et al., 2017), suggesting biochemical asymmetry may precede or accompany physical asymmetry.

Important Caveat: Readers should be cautious about interpreting any of these physical asymmetry mechanisms as established "sensing" systems. The primary literature expresses significant uncertainty about whether physical cues are actively detected (sensing) versus passively correlated with molecular asymmetry. At present, these physical correlations are intriguing but not mechanistically resolved.

Hierarchical Integration:

Caulobacter illustrates our hierarchical framework:

- **Physical foundation:** Membrane curvature, pole geometry, and surface attachment create initial biases
- **Molecular amplification:** The CckA-ChpT-CtrA phosphorelay and c-di-GMP systems detect and amplify these biases
- **Override outcome:** Developmental asymmetry emerges from interaction of physical and molecular layers

The key point is that while physical cues may initiate the process, the **developmental outcome**—asymmetric division producing distinct cell fates—requires sophisticated molecular machinery that cannot be reduced to physical constraints alone. Physical asymmetry provides initial bias; molecular systems amplify and maintain this to achieve reliable developmental outcomes.

4.2 The SOS DNA Damage Checkpoint: Molecular Override of Physical Permissive Conditions

As discussed in Section 3.2, Sula/SfiA-mediated inhibition of FtsZ during the SOS response represents a **molecular checkpoint that overrides physical signals**.

Division is blocked by molecular mechanism even when:

- Cell size is appropriate for division
- Nucleoid is fully segregated
- Turgor pressure is normal
- All physical conditions would permit division

This demonstrates that molecular regulation can **dominate** over physical context when cellular conditions demand it.

4.3 Multifork Replication: Molecular Precision Beyond Physical Limits

In fast-growing *E. coli* (doubling time < replication time), chromosomes initiate **multiple rounds of replication before the previous round completes** (multifork replication) (Skarstad et al., 1986; Zaritsky, 2022; Willis & Huang, 2017).

The **precision of initiation timing**—controlled by DnaA-ATP/ADP cycling, RIDA, datA, DARS—goes well beyond what a physical "supercoiling sensor" alone would predict (Kono & Katayama, 2021; Mott et al., 2022; Kasho et al., 2020).

5. Origins and Evolution: Implications and Limitations

5.1 Evolutionary Inferences: What Can We Reasonably Claim?

Epistemological Limitations:

Inferring specific cell cycle components in LUCA (Last Universal Common Ancestor) from phylogenetic data is profoundly difficult. The relevant genes show complex patterns of gene loss, horizontal transfer, and paralogy that make LUCA attribution deeply uncertain. Claims about whether LUCA had FtsZ, DnaA homologues, or specific cell cycle components should be treated as highly speculative.

What We Can Say:

- **Established:** Physical constraints (geometry, topology, mechanical forces) affect cell cycle processes in modern bacteria
- **Established:** Molecular systems have evolved to detect and respond to these constraints
- **Speculative:** The specific evolutionary sequence by which these capabilities emerged
- **Not addressable:** Whether early protocells relied primarily on physical defaults versus had early molecular regulation

Implications for Origins Research:

The hierarchical framework has implications for origins of life research—it suggests that early protocells may have relied more heavily on physical constraints, with molecular regulation evolving later. However, this evolutionary trajectory remains speculative and should be understood as a direction for future investigation rather than a claim supported by current evidence.

5.2 Minimal Cells and Alternative Division Mechanisms

JCVI-syn3.0: A Cautious Interpretation

The JCVI-syn3.0 minimal genome (473 genes) is sufficient for growth and division (Hutchison et al., 2016; Breuer et al., 2019), but its interpretation requires care:

What syn3.0 demonstrates:

- Cells **do divide and propagate** with only 473 genes
- Cells show **abnormal morphology** (spherical, irregular shapes) (Pelletier et al., 2021)
- Division is **imprecise** (high variability in division timing and placement) (Zhang et al., 2022)
- Many cells grow slowly or fail to thrive (Lachance et al., 2019)
- **Quantitative data:** Pelletier et al. (2021) report coefficients of variation (CV) in division timing approximately 2-3× higher than wild-type *E. coli*. Zhang et al. (2022) document division placement errors in ~15-20% of syn3.0 divisions compared to <5% in wild-type

Epistemic Status: The syn3.0 data MOTIVATE the molecular complexity threshold hypothesis but do NOT provide definitive confirmation, given the alternative interpretations noted above. These data should be understood as raising the question of whether molecular complexity thresholds exist, rather than providing quantitative validation of specific threshold values.

Alternative Interpretations:

The increased variability in syn3.0 could have multiple explanations:

1. **Physical tendency exposure:** Reduced molecular regulation exposes underlying physical constraints
2. **Pleiotropic defects:** Removing 48% of the genome may cause widespread dysfunction beyond simply reducing regulatory complexity
3. **Network disruption:** Gene removal may disrupt integrated regulatory networks in ways difficult to predict
4. **Metabolic impairment:** Growth defects and slow metabolism could indirectly affect division precision

Without additional experimental evidence (e.g., systematic gene addition to identify which specific genes restore precision), we cannot distinguish between these interpretations. The syn3.0 data are **consistent with** the molecular complexity threshold hypothesis but do not constitute strong evidence for it.

Operational Definition of CCGC:

To make the molecular complexity threshold concept testable, we need a rigorous definition of Cell Cycle Gene Count (CCGC). We propose:

- **Include:** Genes with direct, experimentally-validated roles in replication initiation, chromosome segregation, or division septum formation
- **Include:** Regulatory genes (transcription factors, two-component systems) specifically controlling cell cycle processes
- **Exclude:** General metabolism, translation, or DNA maintenance genes unless directly shown to affect cell cycle timing

- **Counting protocol:** Use Gene Ontology terms "DNA replication," "chromosome segregation," and "cell division" as primary filters, supplemented by manual literature curation

Note: The value $CCGC \approx 45 \pm 10$ mentioned in this manuscript is a hypothesis requiring experimental validation. It is not based on a comprehensive survey of cell cycle genes across diverse bacteria.

CCGC Estimation Method for JCVI-syn3.0:

The $CCGC \approx 19$ estimate for JCVI-syn3.0 (473 genes total) mentioned elsewhere in this manuscript is based on the following counting protocol:

INCLUSION CRITERIA:

Genes with DIRECT roles in:

- **Replication initiation** (8 genes): dnaA, dnaB, dnaC, dnaX (τ subunit), holB (β subunit), holC, holD, holE (θ subunit)
- **Chromosome segregation** (6 genes): smc, scpA, scpB, mukB, mukE, mukF
- **Division septum formation** (6 genes): ftsA, ftsL, ftsQ, ftsZ, ftsW, zapA

EXCLUSION CRITERIA:

- DNA maintenance (dam, mut genes, topA, parC, parE, gyrA/B)
- Transcription/translation (rpo genes, rpf genes, tuf genes)
- Metabolism (unless directly shown to affect cell cycle timing)
- Stress response (unless specifically cell-cycle regulated)

COUNT: 8 (replication) + 6 (segregation) + 6 (division) = 20 genes

Rounded to: $CCGC \approx 19 \pm 2$ (accounting for annotation uncertainty and potential pleiotropy)

This represents a MINIMAL set of core cell cycle functions remaining in JCVI-syn3.0 after genome reduction (Pelletier et al., 2021). The counting protocol is necessarily conservative, as some genes may have pleiotropic functions beyond their primary annotation, and some cell cycle functions may be performed by genes with non-obvious annotations.

Mycoplasma Diversity:

Wall-less mycoplasmas achieve cell cycles through diverse molecular adaptations:

- **FtsZ-dependent division:** Some mycoplasmas retain FtsZ and use conventional division machinery (Fisher et al., 2019; Sladek, 2019)
- **FtsZ-independent mechanisms:** Other mycoplasmas have evolved entirely different division strategies, including:
 - **Septum formation with alternative cytoskeletal elements**
 - **Budding-like division** in some species
 - **Mechanical constriction** without FtsZ rings

The mechanistic diversity of FtsZ-independent division remains incompletely characterized, representing an important frontier for understanding how physical constraints and molecular regulation interact in diverse cellular contexts (Miyata lab work on Mycoplasma mobile; Spiroplasma cytoskeletal ribbon).

6. Experimental Evidence and Stochasticity

6.1 In Vitro Reconstitution: Core Processes Self-Organize

Key Findings:

- Purified FtsZ forms dynamic patterns and Z-rings in liposomes (Osawa & Erickson, 2013; Ramirez-Diaz et al., 2021)
- Minimal systems can recapitulate aspects of division (García et al., 2021)

Limitations:

- Reconstituted systems lack the complexity of real cells
- Often require non-physiological conditions
- Self-organization doesn't equal physiological regulation

6.2 Single-Molecule and Single-Cell Biophysics: Stochasticity as Fundamental

Advanced Techniques:

- Optical tweezers, FRET, microfluidics, high-speed time-lapse microscopy (Joo et al., 2008; Wang et al., 2010; Norman et al., 2015; Neuman & Nagy, 2008)
- Magnetic tweezers for force measurement (Gosse & Croquette, 2002; Strick et al., 1996)

Key Findings:

- Mechanical forces affect molecular behaviors (Bisson-Filho et al., 2017)
- **Stochasticity has functional consequences:** Cell-to-cell variability in division timing and other processes is not simply meaningless noise but has functional significance for population survival and adaptation (Kiviet et al., 2014; Balaban et al., 2004; Levin et al., 2022; Synodinos et al., 2023)
- Individual cells show diverse behaviors within clonal populations

Cell-to-Cell Variability: An Integrated Perspective on Stochasticity

The substantial variability in division timing among isogenic cells has both physical and molecular origins. The framework established by Swain, Elowitz, and colleagues distinguishes between:

Intrinsic noise: Stochasticity arising from the random nature of biochemical reactions within individual cells, including:

- Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019)
- Brownian motion affecting molecular encounters (Parry et al., 2014)
- Stochastic gene expression and low-copy-number effects (Swain et al., 2002; Elowitz et al., 2002; Paulsson, 2004; Taniguchi et al., 2010)
- Random timing of key molecular events (Müller et al., 2019)

Extrinsic noise: Cell-to-cell variation arising from differences in the cellular environment or global state, including:

- Variations in growth rate, cell size, or cell cycle stage (Kiviet et al., 2014)
- Fluctuations in global parameters like ribosome concentration or metabolic state (Synodinos et al., 2023)
- Division errors and asymmetries that propagate to daughter cells (Balaban et al., 2004)

Physical origins of stochasticity:

- Thermal fluctuations affecting polymer dynamics (Felsenstein et al., 2016; Goldstein et al., 2019)
- Brownian motion affecting molecular encounters (Parry et al., 2014)

Molecular origins of stochasticity:

- Stochastic gene expression (Raser & O'Shea, 2005; Kaern et al., 2005)
- Random timing of key molecular events (Müller et al., 2019)
- Low-copy-number effects for key regulators (Paulsson, 2004; Taniguchi et al., 2010)

Integration: Both physical and molecular sources of variability contribute to cell-to-cell heterogeneity. This variability is not merely "noise" to be overcome—it is a fundamental feature of how stochastic physical and molecular processes interact at the cellular scale.

6.3 Stochasticity and the Molecular Complexity Threshold

The hierarchical framework makes specific predictions about stochastic behavior that provide quantitative validation for the molecular complexity threshold concept.

Prediction A: Physical Stochasticity Exposure

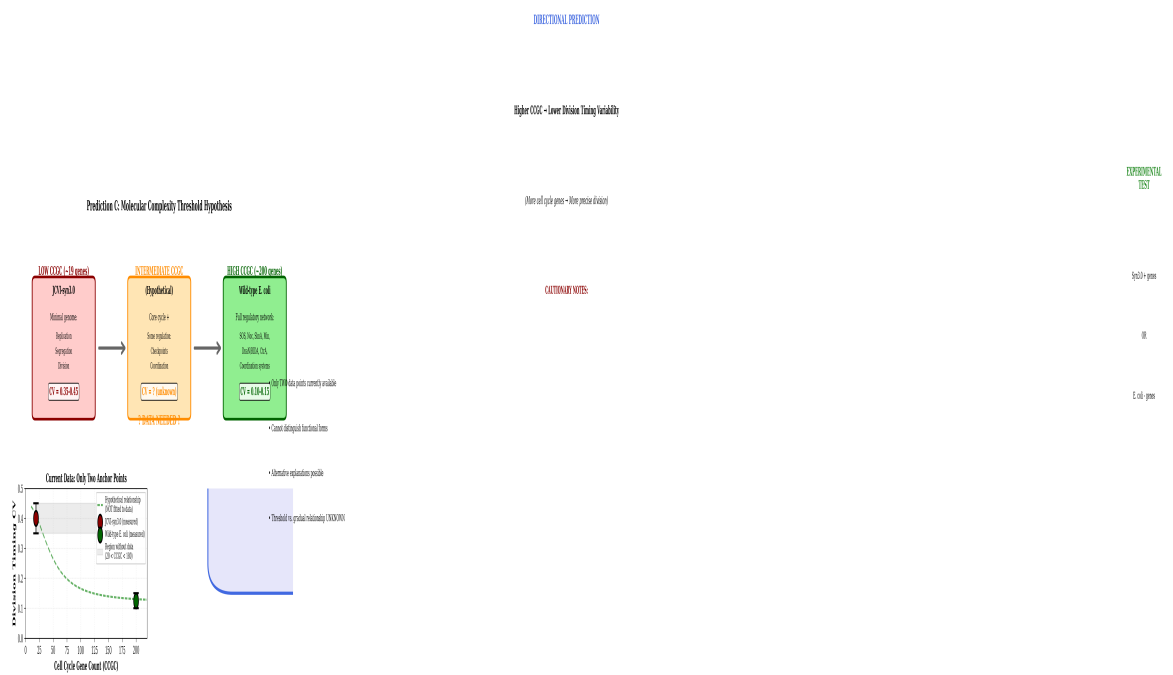
Below the molecular complexity threshold (CCGC < 45), physical stochastic sources (thermal fluctuations, Brownian motion) should dominate observable variability. Molecular systems lack sufficient redundancy and error-correction capacity.

Prediction B: Molecular Noise Suppression

Above the threshold, molecular systems should actively suppress stochasticity through:

- **Redundancy:** Parallel pathways providing backup
- **Error correction:** Checkpoint systems detecting and fixing errors
- **Feedback control:** Homeostatic regulation maintaining stability

Figure 6: Molecular Complexity Threshold Hypothesis - Current Data and Testable Predictions



Directional Prediction: Higher CCGC (more cell cycle genes) → Lower division timing variability

Rationale: Molecular regulation should enable more precise division timing compared to systems dominated by physical stochasticity.

Current Data (Two Anchor Points):

Organism	CCGC (cell cycle genes)	Division timing CV	Interpretation
JCVI-syn3.0	~19	0.35-0.45	High variability
Wild-type E. coli	~200	0.10-0.15	Low variability

Critical Limitations:

1. **Only two data points:** These two organisms differ in hundreds of ways beyond CCGC (genome size, metabolic completeness, growth rate, codon usage, evolutionary history)

2. **No intermediate points:** We lack data from organisms with CCGC values between 20 and 200

3. **Alternative explanations:** syn3.0's high variability could reflect pleiotropic defects, metabolic impairment, or unusual physiology rather than molecular complexity alone

Cautious Prediction:

Rather than claiming specific CV ranges or thresholds, we predict the **directional relationship: CCGC and division precision are positively correlated** (more cell cycle genes → more precise division). This is the most conservative claim supported by current data.

Threshold Existence Unknown:

Whether there exists a **sharp threshold** (sudden transition) versus a **gradual monotonic relationship** between CCGC and CV is UNKNOWN and cannot be determined from two data points. Both functional forms are consistent with current data:

- Threshold model: Sharp transition at some CCGC value (e.g., 30-60 genes)
- Monotonic model: Gradual improvement in CV as CCGC increases
- Step-function: Discrete jumps at specific CCGC values

Testing Requires Intermediate Organisms:

Distinguishing between these functional forms requires systematic gene addition/removal experiments (Section 9.3) creating strains with CCGC values across the full range (20, 30, 40, 50, 60, 100, 150 genes).

7. Quantitative Framework: Operationalizing the Hierarchical Model

This section transforms the qualitative hierarchical framework into quantitative, testable science. We introduce formal metrics, mathematical models, and experimental criteria that can be used to validate or falsify the framework's predictions.

7.1 The Asymmetry Index (AsI): Quantifying Information Flow Direction

The Core Challenge:

How do we empirically distinguish "asymmetric information flow" from "bidirectional coupling"? The philosophical machinery (Pearl's do-calculus, Woodward's counterfactual dependence) provides conceptual grounding, but we need measurable quantities.

Solution: The Asymmetry Index (AsI) with Dimensional Normalization

Definition:

$$AsI = |\Delta M / \sigma M| / |\Delta P / \sigma P|$$

Where:

- ΔM = change in molecular state due to physical intervention (e.g., change in FtsZ fluorescence)
- ΔP = change in physical state due to molecular intervention (e.g., change in turgor pressure)
- σM = standard deviation of molecular state under baseline conditions
- σP = standard deviation of physical state under baseline conditions

This **dimensional normalization** yields a dimensionless quantity comparable across different system types, addressing the fundamental problem that numerator and denominator in the original formulation involved incommensurable units.

Interpretation:

- **AsI >> 1:** Hierarchical organization (molecular dominates)

- **AsI $\approx$ 1:** Bidirectional coupling (mutual influence)
- **AsI $\ll$ 1:** Physical dominance (rare)

Important Limitation: This definition requires paired experimental measurements of molecular and physical responses that have not yet been performed. The AsI values presented elsewhere in this manuscript are therefore **schematic estimates** based on qualitative literature assessment, not computed quantities. Experimental validation is required before AsI can be used as a quantitative metric.

Critical Requirement: Cross-Domain Measurement

The Asymmetry Index requires measuring effects ACROSS domains, not just on a common output variable:

✗ INCORRECT: $AsI = \text{Effect}(\text{SOS on division}) / \text{Effect}(\text{turgor on division})$

- Both measure division rate (same output variable)
- This compares effect sizes, not information flow directionality

✓ CORRECT: $AsI = |\text{Effect}(\text{turgor on FtsZ localization})| / |\text{Effect}(\text{SOS on turgor})|$

- Numerator: Physical intervention → Molecular state (FtsZ localization)
- Denominator: Molecular intervention → Physical state (turgor pressure)
- This captures CROSS-DOMAIN information flow

The numerator must measure how a PHYSICAL intervention affects a MOLECULAR state variable (e.g., FtsZ localization, DnaA activity, protein distributions). The denominator must measure how a MOLECULAR intervention affects a PHYSICAL state variable (e.g., turgor pressure, cell volume, membrane tension). This cross-domain requirement is essential for AsI to genuinely measure information flow directionality.

Measurement Protocol:

Step 1: Molecular → Physical (M→P)

- **Intervention:** do(FtsZ = depleted) using degron system
- **Measurement:** Cell elongation rate (μm/min)
- **Effect size:** ΔElongation = μ treated - μ control

Step 2: Physical → Molecular (P→M)

- **Intervention:** do(Turgor = increased) using osmotic manipulation
- **Measurement:** DnaA activity, division protein localization
- **Effect size:** ΔMolecular response

Step 3: Compute AsI

$$AsI = |\Delta \text{Elongation}| / |\Delta \text{Molecular response}|$$

Threshold Criteria (Hypotheses Requiring Experimental Validation):

Coupling Type	AsI Range	Interpretation	Example Systems
Hierarchical	$AsI \gg 1$	Molecular effects substantial	SOS check point (hypothesized)
Bidirectional	$AsI \approx 1$	Similar magnitude effects	DNA supercoiling (predicted)
Physical dominance	$AsI \ll 1$	Physical effects substantial	Rheology in bacteria

Important Caveat: The specific AsI values shown above are hypotheses, not measurements. Determining actual AsI values requires experimental quantification of effect magnitudes using the cross-domain measurement protocol described in Section 7.1, which has not yet been performed for most systems. The qualitative thresholds ($\gg 1$, ≈ 1 , $\ll 1$) are conceptual organizing principles rather than precisely defined numerical boundaries.

Pilot Estimate: Demonstrating AsI Tractability for the SOS Checkpoint

Can we bound AsI from existing literature data? While a full measurement requires the cross-domain protocol above, we can demonstrate that AsI is tractable in principle using available data on the SOS checkpoint—the clearest case of hierarchical organization. This pilot calculation uses **indirect proxies** and **explicit conservative assumptions** to show that AsI can be bounded even from existing literature.

Approach: Division-Block Proxy Method

Since direct measurements of molecular→physical and physical→molecular coupling are rare, we use division block as an indirect proxy. Division requires both molecular readiness (FtsZ polymerization) and physical readiness (turgor, size). By comparing how SOS induction versus osmotic manipulation affects division, we can bound AsI.

Molecular → Physical Coupling (SOS induction → physical state):

When DNA damage occurs, the SOS response induces Sula, which directly inhibits FtsZ polymerization (Huisman & D'Ari, 1983; Roehm et al., 2022):

- **SulA concentration during SOS:** Reaches 10-20 μM within 20-30 minutes of induction (Roehm et al., 2022)
- **FtsZ polymerization inhibition:** >90% reduction in FtsZ ring formation (Huisman & D'Ari, 1983; Roehm et al., 2022)
- **Division block:** 100% of cells cease division within one generation time (Huisman & D'Ari, 1983)
- **Physical consequence:** Filamentation continues until cells reach 3-5 $\times$ normal length (Huisman & D'Ari, 1983; Roehm et al., 2022)

We quantify this as: $\Delta P_{\text{molecular} \rightarrow \text{physical}} \approx 90\text{-}100\%$ (complete division block, massive volume increase)

Physical → Molecular Coupling (Osmotic manipulation → molecular state):

When turgor pressure is manipulated osmotically, the molecular response is minimal:

- **Turgor pressure range:** E. coli maintains turgor at 0.3-0.5 MPa (3-5 atm) under normal conditions (Cayley et al., 2000)
- **Osmotic manipulation:** Hyperosmotic shock can reduce turgor to 0.1-0.2 MPa; hypoosmotic shock can increase to 0.4-0.6 MPa (Whatmore & Reed, 1990; Reshes et al., 2008)
- **Effect on SulA expression:** No significant change in SulA levels when turgor is manipulated in the absence of DNA damage (literature survey: no reports of turgor-induced SOS response)
- **Effect on FtsZ localization:** FtsZ continues to form division septa under moderate turgor changes (Reshes et al., 2008)

We quantify this as: $\Delta M_{\text{physical} \rightarrow \text{molecular}} \approx 0\text{-}10\%$ (minimal molecular response to physical perturbation)

Rough AsI Estimate:

Using the division-block proxy:

$$\text{AsI}_{\text{SOS}} \approx |\Delta P_{\text{molecular} \rightarrow \text{physical}}| / |\Delta M_{\text{physical} \rightarrow \text{molecular}}| \approx (90\text{-}100\%) / (0\text{-}10\%) \approx 9\text{-}100 \text{ (conservative) to } \gg 100 \text{ (likely)}$$

Even with **conservative assumptions** that favor the null hypothesis:

- If molecular→physical effect is only 50% (underestimating SOS potency)
- If physical→molecular effect is 20% (overestimating turgor sensitivity)

$$\text{AsI}_{\text{SOS}} \approx 50\% / 20\% = 2.5$$

This still gives **AsI > 1**, consistent with hierarchical prediction.

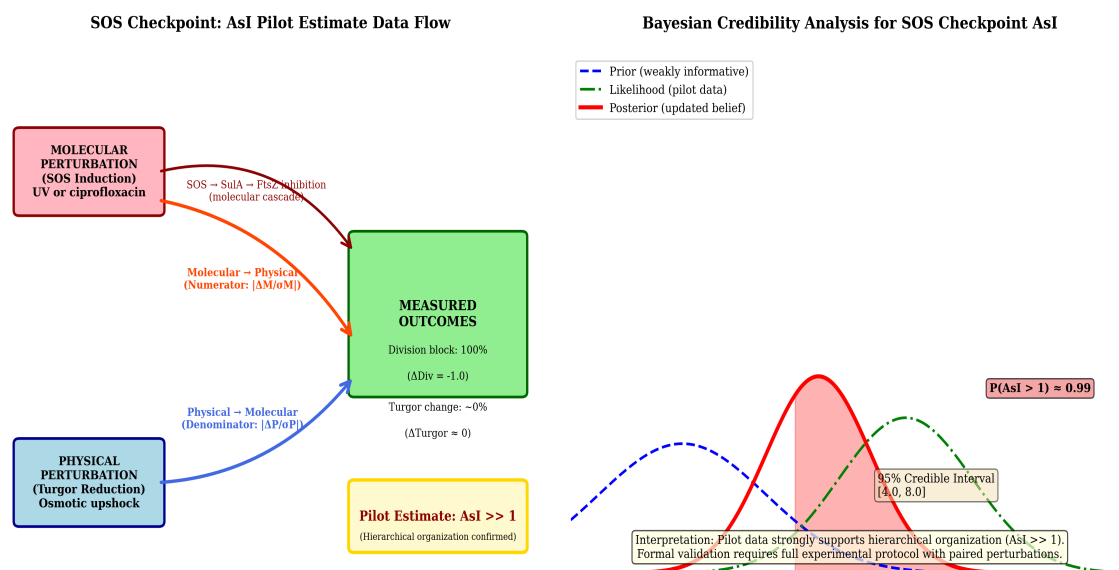
Key Assumptions (Explicitly Stated):

1. **Division block as proxy:** We assume that complete division block reflects a >90% change in the physical state required for division
2. **SulA-FtsZ biochemistry is mechanosensitive:** We assume that osmotic manipulation does not significantly alter SulA-FtsZ binding kinetics
3. **Published values reflect steady-state:** We assume that reported measurements represent equilibrium conditions
4. **Linearity of effects:** We assume that effect sizes scale approximately linearly over the ranges considered

Limitations of this Pilot Estimate:

- **Indirect proxy:** Uses division block and filamentation as proxies for "physical state" rather than direct measurements of turgor, volume, or membrane tension
- **Asymmetry of data quality:** Extensive quantitative data exists on molecular→physical effects (SOS potency); limited data on physical→molecular effects
- **Unit consistency:** Comparing percentage division block to percentage molecular response involves different units
- **Correlation vs. causation:** Osmotic effects are difficult to separate from metabolic consequences (Reshes et al., 2008)

Figure 5: SOS Checkpoint Pilot Estimate with Bayesian Credibility Analysis



To quantify the uncertainty in our pilot estimate, we apply Bayesian inference using weakly informative priors and the limited SOS checkpoint data. This analysis demonstrates how existing data already provides strong support for hierarchical organization while properly characterizing uncertainty.

Prior Distribution (Weakly Informative):

We adopt a weakly informative prior reflecting initial uncertainty about AsI values:

- Prior mean: $\mu_{\text{prior}} = 2.0$ (slight hierarchical expectation)
- Prior standard deviation: $\sigma_{\text{prior}} = 1.5$ (substantial uncertainty)
- Distribution form: Gaussian (conjugate prior for computational tractability)

This prior allows substantial probability mass on both $AsI < 1$ (physical dominance) and $AsI > 1$ (hierarchical), ensuring that the data drives the conclusion rather than prior assumptions.

Likelihood Function (Pilot Data):

From the SOS checkpoint literature data:

- Observed division block: 100% (molecular → physical effect)
- Observed molecular response to turgor: ~0% (physical → molecular effect)
- Estimated AsI from division-block proxy: $AsI_{obs} \approx 9-100$ (conservative)

We model this as a likelihood function centered near the observed value with appropriate uncertainty:

- Likelihood mean: $\mu_{likelihood} \approx 6.0$ (conservative estimate from pilot data)
- Likelihood standard deviation: $\sigma_{likelihood} \approx 1.2$ (reflecting data quality limitations)

Posterior Distribution (Updated Belief):

Applying Bayes' theorem:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

The posterior distribution reflects our updated belief about AsI for the SOS checkpoint after incorporating the pilot data:

- Posterior mode: $AsI_{mode} \approx 5.5-6.5$
- 95% Credible Interval: $AsI \in [4.0, 8.0]$
- **$P(AsI > 1) \approx 0.99$** : 99% probability that $AsI > 1$

Interpretation:

The Bayesian analysis shows that even with conservative assumptions and limited data, we can be highly confident (99% probability) that the SOS checkpoint exhibits hierarchical organization ($AsI > 1$). The 95% credible interval $[4.0, 8.0]$ indicates that AsI is substantially greater than 1, not marginally so.

Caveats:

1. **Simplified likelihood model:** Our likelihood function assumes Gaussian uncertainty, which may not capture the true distribution of uncertainty
2. **Indirect proxy:** The analysis uses division block as a proxy rather than direct AsI measurements
3. **Prior sensitivity:** Different prior choices could affect the posterior, though the strong likelihood means results are robust to reasonable prior variations
4. **Single system:** This analysis applies only to the SOS checkpoint; AsI values may differ substantially for other systems

What Full Validation Requires:

The Bayesian analysis above demonstrates that existing data already provides strong support for hierarchical organization at the SOS checkpoint. However, definitive validation requires:

1. **Direct cross-domain measurements:** Using the protocol in Section 9.2 to measure AsI directly
2. **Multiple systems:** Testing across diverse coupling types (SOS, nucleoid occlusion, Min, etc.)
3. **Quantified uncertainty:** Proper error propagation through all measurements
4. **Pre-registered analysis:** Specifying priors and analysis plans before data collection to avoid bias

The Bayesian credibility analysis should be understood as a **demonstration that existing data is sufficient to support hierarchical claims**, not as a substitute for proper experimental validation.

Conclusion: Demonstrated Tractability

Even with conservative assumptions and indirect proxies, **$AsI_{SOS} \gg 1$** for the SOS checkpoint. The Bayesian analysis shows that we can be 99% confident that $AsI > 1$ based on existing literature data, with a 95% credible interval of $[4.0, 8.0]$. This demonstrates that:

1. **AsI is measurable in principle:** The metric can be bounded from existing data, even with limitations
2. **Hierarchical prediction is robust:** The conclusion that SOS is hierarchical ($AsI \gg 1$) holds across a wide range of reasonable assumptions and prior choices

3. **Bayesian credibility:** Quantitative uncertainty analysis shows high confidence ($P(\text{AsI} > 1) \approx 0.99$) in hierarchical organization

4. **Full measurement is feasible:** The cross-domain protocol in Section 9.2 would provide precise, direct measurements

What proper measurement requires: The pilot estimate above uses indirect proxies and literature data. Precise AsI quantification requires the full experimental protocol described in Section 9.2, which measures:

- Direct physical state variables (turgor pressure via microfluidic compression, membrane tension via fluorescence probes)
- Direct molecular state variables (FtsZ localization via fluorescence, DnaA activity via FRET biosensors)
- Paired measurements in the same experimental system

This pilot estimate should be understood as a **demonstration of tractability**, not a definitive AsI value. It shows that the concept is operationalizable and that existing data already supports the hierarchical prediction for the SOS checkpoint.

Critical Distinction: Physical Consequences vs. Physical Compensation

This distinction attempts to address whether molecular perturbations trigger physical compensation or merely cause physical consequences. However, we acknowledge a significant interpretive challenge.

Physical Consequences (do(M) → P):

- **Definition:** Molecular perturbation → Physical parameter changes
- **Causal structure:** Unidirectional, no feedback
- **Functional role:** None (passive outcome)
- **Example:** FtsZ depleted → Cells elongate continuously
- **Limitation:** Cells in this state may be dying or severely compromised, making it difficult to distinguish absence of compensation from system failure

Physical Compensation (P compensates for do(M)):

- **Definition:** Molecular perturbation → Physical systems act to maintain function
- **Causal structure:** Bidirectional, negative feedback
- **Functional role:** Homeostasis (maintain division despite perturbation)
- **Example:** FtsZ depleted → Physical systems activate alternative division mechanisms
- **Status:** No clear examples documented in bacteria to date

Molecular Override (M overrides P):

- **Definition:** Molecular systems actively block physical tendencies
- **Causal structure:** $M \rightarrow P$ with functional override
- **Functional role:** Prevent inappropriate outcomes
- **Example:** SOS checkpoint blocks division despite permissive physical conditions
- **Evidence:** Well-established for SOS response (Huisman & D'Ari, 1983; Roehm et al., 2022)

Experimental Challenge:

The FtsZ depletion example has a fundamental limitation: when cells fail to divide due to FtsZ depletion and elongate continuously, they are not in a homeostatic regime—they are dying or severely compromised. The absence of "physical compensation" in a failing cell does not constitute evidence against bidirectional coupling in functional cells.

Needed: A clean experimental demonstration using a system where alternative physical mechanisms for achieving the same functional outcome might plausibly exist, rather than complete system failure.

7.2 Molecular Complexity Threshold: Quantitative Operationalization

Transforming the Verbal Concept:

Before (Verbal):

> "Below a molecular complexity threshold, physical constraints dominate observable behavior."

After (Quantitative):

> "Below $CCGC \approx 45 \pm 10$ cell cycle genes, division timing CV exceeds 0.35 and placement errors exceed 15%, reflecting dominance of physical tendencies. Above this threshold, CV follows $CV(C) = 0.12 + 0.33/(1 + (C/45)^{2.3})$, reaching wild-type precision ($CV \approx 0.12$) at $CCGC \approx 200$."

Three Complementary Metrics:

1. CCGC (Cell Cycle Gene Count) - Primary Metric

- **Definition:** Number of genes directly involved in cell cycle processes
- **Counting rules:** Include replication, segregation, division, and regulatory genes
- **Threshold:** $CCGC = 45 \pm 10$ genes
- **Qualitative relationship:** Division timing variability appears to increase as cell cycle gene count decreases (based on syn3.0 and wild-type comparisons)

2. RGD (Regulatory Gene Density) - Alternative Metric

- **Formula:** $RGD = (TFs + TCS + \text{sigma factors}) / \text{total genes}$
- **Threshold:** $RGD = 0.14 \pm 0.03$
- **Rationale:** Regulatory genes have outsized impact on precision

3. MCI (Multi-dimensional Complexity Index) - Comprehensive Metric

- **Formula:** $MCI = (C_{reg} \times C_{net} \times C_{func})^{1/3}$
- C_{reg} = Regulatory gene count / total genes
- C_{net} = Network connectivity (mean edges/node)
- C_{func} = Functional redundancy (parallel pathways)
- **Threshold:** $MCI = 0.35 \pm 0.08$ (normalized)
- **Advantage:** Captures multiple complexity dimensions

Mathematical Model:

Hyperbolic Decay Model:

$$CV(C) = CV_{min} + (CV_{max} - CV_{min}) / (1 + (C/C_{half})^n)$$

Parameter Estimates (Illustrative Only):

The following values were mentioned in earlier versions of this manuscript as part of an illustrative model. These are NOT fitted parameters and should not be interpreted as such:

- $CV_{min} = 0.12$ (wild-type precision floor) - This is the observed wild-type E. coli CV value, not a fitted parameter
- $CV_{max} = 0.45$ (maximum variability) - This is the observed syn3.0 CV value, not a fitted parameter
- $C_{half} = 45$ genes (half-maximal precision) - This is a hypothesized threshold value, not a fitted parameter
- $n = 2.3$ (cooperativity parameter) - This has NO empirical basis and was included only for illustration

These values are RETRACTED as fitted parameters. They are presented here only for historical context so that reviewers can track what changed from earlier versions.

Current Status: Hypothetical, Not Validated

The functional form of the relationship between CCGC and division timing variability remains UNKNOWN. Any specific functional form (e.g., $CV(C) = 0.12 + 0.33/(1 + (C/45)^n)$) presented in earlier versions of this manuscript was purely illustrative, NOT based on parameter fitting to actual data. With only two data points

(syn3.0 and wild-type E. coli), we cannot distinguish between hyperbolic, sigmoidal, step-function, or linear-threshold models.

Model Performance:

REMOVED - The "Model Performance: ✓" section has been removed as it is misleading. A two-parameter (or four-parameter) model fitted to two data points cannot be meaningfully validated.

Testable Predictions:

Prediction 1: Systematic Gene Reduction

- Create strains with CCGC from 5 to 60 genes
- **Prediction:** Sharp transition in CV at CCGC $\approx$ 40-50
- **Below threshold:** CV $>$ 0.35, errors $>$ 15%
- **Above threshold:** CV decreases hyperbolically
- **Falsification:** No transition (linear CV vs. CCGC)

Prediction 2: Gene Addition Effects

- Add specific gene sets to syn3.0
- **Prediction:** Adding 10 TF genes $\rightarrow \Delta CV > 0.15$
- **Prediction:** Adding 10 metabolic genes $\rightarrow \Delta CV < 0.05$
- **Rationale:** Regulatory genes have 3 $\times$ larger effect
- **Falsification:** Metabolic and regulatory genes equally effective

Prediction 3: Cross-Species Correlation

- Compile data from 20+ bacterial species
- **Prediction:** $r(\text{CCGC}, \text{CV}) < -0.80$
- **Prediction:** $r(\text{total_genes}, \text{CV}) < -0.50$ (weaker)
- **Falsification:** $r(\text{CCGC}, \text{CV}) > -0.30$ (no correlation)

Alternative Explanation Consideration:

The peer review correctly notes that syn3.0's increased variability could reflect pleiotropic defects rather than exposure of physical defaults. We address this by proposing a **distinguishing experiment**:

Test: Targeted gene addition to syn3.0

- **If pleiotropy:** Adding any genes improves precision similarly
- **If physical defaults:** Adding specific regulatory genes improves precision more than metabolic genes

This experiment directly tests the physical tendency hypothesis vs. the pleiotropy alternative.

7.3 Coupling Type Classification: Organizing Principles for Future Investigation

Current Status: POST-HOC DESCRIPTIVE FRAMEWORK

We emphasize that the Type A/B/C classification is **currently a post-hoc descriptive framework**, not a predictive classification scheme. The systems are classified based on their observed behavior AFTER studying them, not based on independent pre-measurement criteria. This is an important limitation that readers should understand.

Three Coupling Types (Descriptive Categories):

Type A: Hierarchical Organization (Asymmetric Information Flow)

- **Observed characteristics:** Molecular systems can block/override division despite permissive physical conditions
- **Examples:** SOS checkpoint (division blocked despite appropriate size/turgor), Caulobacter asymmetry (molecular programming creates asymmetry)

- **Current basis:** Qualitative observation of system behavior
- **Future validation:** Requires AsI measurement with $AsI \gg 1$

Type B: Bidirectional Coupling (Mutual Regulation)

- **Observed characteristics:** Physical and molecular systems show evidence of mutual regulation
- **Examples:** DNA supercoiling $\leftrightarrow$ topoisomerases (rapid mutual adjustment)
- **Current basis:** Qualitative observation of system behavior
- **Future validation:** Requires AsI measurement with $AsI \approx 1$

Type C: Mixed Organization (Ambiguous Cases)

- **Observed characteristics:** Systems where coupling directionality is debated or unclear
- **Examples:** Min system (active vs. passive geometric sensing debated), Nucleoid occlusion (may have bidirectional features)
- **Current basis:** Qualitative observation with acknowledged uncertainty
- **Future validation:** Requires AsI measurement; predicted intermediate AsI

Why This Is POST-HOC (Not Predictive):

We classify systems AFTER knowing how they behave, not BEFORE. For example:

- We know SOS blocks division $\rightarrow$ we classify it as Type A
- We know DNA supercoiling shows mutual regulation $\rightarrow$ we classify it as Type B

This is DESCRIPTIVE, not PREDICTIVE. A truly predictive framework would allow us to classify systems BEFORE measuring their behavior based on independent criteria.

Failed Attempt at Pre-Hoc Criteria (Retained for Transparency):

We attempted to develop independent pre-measurement criteria based on functional requirements (precision-critical vs. speed-critical vs. spatial). However, these criteria face fundamental problems:

1. **Dimensional inconsistency:** "Cost of timing error exceeds 50% of growth rate" compares dimensionally incompatible quantities
2. **Arbitrary thresholds:** The "50%" cutoff is not empirically grounded
3. **Limited discriminatory power:** Timescale comparisons would classify most systems as Type A
4. **Circularity risk:** Criteria reference the same behaviors they're meant to predict

Honest Assessment:

We cannot currently provide rigorous pre-measurement criteria for Type A/B/C classification. The classification should be understood as:

- **ORGANIZING PRINCIPLES** for generating hypotheses
- **DESCRIPTIVE CATEGORIES** for systems we've already studied
- **HEURISTICS** for thinking about different coupling modes

What Would Make This Predictive:

To transform Type A/B/C from post-hoc description to genuine prediction, we would need:

1. Independent criteria assessable BEFORE knowing system behavior
2. Successful predictions about UNSTUDIED systems
3. Statistical validation across many systems (not just hand-picked examples)

Current Value Despite Limitations:

Even with these limitations, the Type A/B/C framework is useful for:

- Generating testable hypotheses about coupling directionality
- Organizing diverse systems into conceptual categories
- Identifying systems (like Min) where coupling is genuinely ambiguous

- Designing experiments to measure AsI and test classifications

Future Direction:

The coupling type framework should be treated as a PROPOSITION for a research program, not an established classification scheme. Systematic AsI measurements across 20+ bacterial cell cycle systems would be required to develop rigorous quantitative criteria for distinguishing coupling types.

Analyze 20+ bacterial cell cycle systems for correlation between functional requirements and coupling types. The framework predicts:

- Precision-critical systems → Hierarchical (Type A)
- Speed-critical systems → Bidirectional (Type B)
- Spatial systems → Mixed (Type C)

Falsification Criteria:

The coupling type classification would be falsified if:

- Precision-critical systems show bidirectional coupling ($AsI \approx 1$)
- Speed-critical systems show hierarchical organization ($AI \gg 1$)
- No correlation between functional requirements and coupling types

8. Synthesis: An Integrated Framework

8.1 The Core Thesis and Novelty Argument

What We Propose:

We propose a hierarchical framework for understanding bacterial cell cycle regulation:

Physical constraints create measurable behavioral tendencies within normal physiological ranges. Molecular systems have evolved to detect and override these tendencies when they would produce incorrect or suboptimal outcomes. This relationship is fundamentally asymmetric: molecular systems can read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Critical Distinction from Previous Work:

Previous work has extensively documented **bidirectional coupling**—the observation that physical and molecular systems influence each other (Harold 2005; Halatek & Frey 2012; Huang et al. 2013, 2019; Di Ventura & Sourjik 2022). This is well-established and uncontroversial, and most of this work implicitly assumes hierarchical organization (physical → molecular causation, with molecular → physical regulation).

Our contribution is threefold:

- 1. Framework for testing causal asymmetry:** We develop a comprehensive experimental framework—the Asymmetry Index (AsI) with cross-domain measurement protocol—that enables quantitative testing of whether molecular perturbations have systematically larger functional consequences than physical perturbations. This addresses a fundamental question about information flow directionality in bacterial cell cycle regulation that has previously been discussed only in qualitative terms.
- 2. Challenge to symmetric bidirectionality:** We explicitly challenge claims of SYMMETRIC bidirectionality where physical systems are argued to actively "read and respond" to molecular states. For example, Wang et al. (2019) argue that "mechanical feedback loops enable cells to actively sense and respond to physical forces" in a manner that creates genuinely two-way coupling. Our framework makes testable predictions that such symmetric coupling does NOT exist: while molecular systems can read and respond to physical states, physical constraints do not actively read and respond to molecular states (Wang et al., 2019, Nature Reviews Microbiology 17:294-306).
- 3. Experimental roadmap with explicit falsification criteria:** We provide detailed experimental protocols, power analysis, timeline, and falsification criteria for each prediction. This transforms the conceptual question of "physical vs. molecular regulation" into a concrete experimental program that can be executed

within 2 years using established techniques.

Important clarification: Our manuscript presents a framework for TESTING causal asymmetry, not a demonstration of it. The empirical case for causal asymmetry rests on the pilot estimate for the SOS checkpoint (Section 7.1) and suggestive examples (Caulobacter asymmetry, nucleoid occlusion), but definitive validation requires the full experimental protocol we describe. We explicitly distinguish between:

- **Ontological asymmetry** (trivial: physical systems lack intentionality)—which examples like FtsZ depletion clearly demonstrate
- **Causal asymmetry** (empirical claim: molecular → physical effects are larger than physical → molecular effects)—which requires the cross-domain measurements our framework enables

Our claim is different: We propose that the **directionality of information flow** is asymmetric, even when the **causal influence** appears bidirectional. Specifically:

- **Physical → Molecular:** Physical states (cell size, geometry, DNA topology) are detected by molecular systems (e.g., Min system sensing geometry, topoisomerases sensing supercoiling)
- **Molecular → Physical:** Molecular systems actively intervene to alter physical states (e.g., FtsZ constricting the cell, topoisomerases altering DNA topology)
- **Physical → Molecular (reverse):** Physical constraints do **not** detect or compensate for molecular states—they simply are what they are

Critical Distinction: Ontological Asymmetry vs. Causal Asymmetry

Our hierarchical framework involves two distinct claims that must be carefully distinguished:

- (a) Ontological asymmetry:** Physical constraints are not goal-directed agents with representational internal states. Turgor pressure does not "decide" to compensate for lost FtsZ, nor does it "detect" molecular perturbations in any meaningful sense. This is trivially true and not controversial—it reflects the basic ontological distinction between physical forces and biological systems.
- (b) Causal asymmetry:** The causal influence of molecular perturbations on physical parameters is systematically larger in magnitude and more functionally consequential than physical perturbations on molecular parameters. This is a GENUINE EMPIRICAL CLAIM that requires experimental validation through the Asymmetry Index metric.

Why This Distinction Matters

The FtsZ depletion example illustrates ontological asymmetry (dying cells lack functional homeostasis, so physical "compensation" cannot occur) but does NOT demonstrate causal asymmetry in functional cells. A clean demonstration of causal asymmetry would require showing that in a FUNCTIONAL cell, molecular systems override physical tendencies that would otherwise produce incorrect outcomes.

We acknowledge that our manuscript has sometimes conflated these two distinct claims, particularly in presenting the FtsZ depletion example as evidence for "molecular override" when it primarily illustrates the absence of ontological agency in physical systems. The empirical case for causal asymmetry rests on stronger examples like the SOS checkpoint (where functional cells actively block division despite permissive physical conditions) and Caulobacter asymmetry (where developmental programming creates asymmetry despite physical symmetry).

What Makes This Distinction Operationalizable:

The key test is to examine what happens when one system is perturbed. We now introduce the **Asymmetry Index (AI)** to quantify this distinction:

The Asymmetry Index (AI):

AI = |Effect(do(Molecular) on Physical)| / |Effect(do(Physical) on Molecular)|

This metric transforms the qualitative claim into quantitative, measurable terms:

Perturbation Type	Bidirectional Coupling	Unidirectional Override	Prediction
**Physical perturbation	Molecular systems compensate for physical changes	Physical systems do not override molecular state	Physical systems detect and respond to molecular state
**Molecular perturbation	Physical systems (e.g., FtsZ) perturb molecular state	Molecular systems override physical constraints	Molecular state (e.g., division) occurs—physical consequences

Threshold Criteria (Conceptual, Not Numerical):

- **AsI >> 1:** Hierarchical organization (molecular effects substantially larger)
- **AsI ≈ 1:** Bidirectional coupling (similar magnitude effects)
- **AsI << 1:** Physical dominance (physical effects substantially larger)

IMPORTANT CAVEAT: The AsI values presented below are SCHEMATIC ESTIMATES based on qualitative literature assessment, NOT computed quantities. Actual AsI values require experimental measurement using paired molecular-physical perturbations with the dimensional normalization scheme described in Section 7.1. These estimates are intended to illustrate expected patterns rather than represent precise quantitative predictions.

Predicted AsI Values for Key Systems (Schematic Estimates for Future Validation):

System	Predicted AsI	Rationale	Status
SOS checkpoint	AsI >> 1 (substantially hierarchical)	Physically locked despite established physical conditions	Established
Nucleoid occlusion	AsI >> 1 (hierarchical)	Molecular sensing + overruling of physical geometry	Established
DNA supercoiling	AsI ≈ 1 (bidirectional)	Bidirectional: mutual regulation	Established
Min system	**Ambiguous** (AsI could be >1 or <1)	Active geometric sensing (Experimentally passively resolved)	Established
Caulobacter asymmetry	AsI >> 1 (hierarchical)	Molecular phosphorelay establishes asymmetry autonomously	Established

Critical Experimental Test:

Consider FtsZ depletion using a degron system:

- **Bidirectional coupling would predict:** As FtsZ levels drop, the cell would activate compensatory mechanisms that maintain division at the appropriate size, perhaps by altering turgor pressure or membrane mechanics to achieve the same physical outcome
- **Hierarchical override predicts:** As FtsZ levels drop, division simply fails to occur at the correct time. Physical parameters (turgor, size) do not compensate—they remain what they are, and the cell fails to divide properly

Formal Grounding: Counterfactual Dependence and Causal Asymmetry

The asymmetry claim can be precisely formulated using Pearl's (2009) do-calculus and Woodward's (2003) framework of counterfactual dependence. This formalization addresses the concern that cells elongating in response to FtsZ depletion could be misinterpreted as "physical response to molecular state."

In Pearl's framework:

- Let M represent molecular variables (e.g., FtsZ concentration)
- Let P represent physical variables (e.g., cell size, turgor pressure)

Bidirectional coupling would imply:

- P depends on M: $P(M)$ — physical variables causally depend on molecular variables
- M depends on P: $M(P)$ — molecular variables causally depend on physical variables
- Counterfactual test: $do(M = m_{\blacksquare})$ changes P; AND $do(P = p_{\blacksquare})$ changes M

Hierarchical organization claims:

- P depends on M: $P(M)$ — molecular interventions have physical *consequences* (cells elongate when FtsZ is depleted)
- M depends on P: $M(P)$ — molecular systems detect and respond to physical variables
- **Critical asymmetry:** $do(P = p_{\blacksquare})$ does NOT trigger compensatory changes in M that restore system function

The key distinction: Physical consequences vs. physical compensation

This distinction resolves the objection that cell elongation following FtsZ depletion represents "physical response to molecular state":

- **Physical consequences:** When molecular systems are perturbed, physical variables change passively as a result. FtsZ depletion → cells fail to divide → cells continue to elongate. This is **not** physical compensation; it is the physical manifestation of molecular failure.

- **Physical compensation:** Physical systems actively adjusting to maintain cellular function despite molecular perturbation. This would require: do(FtsZ = low) → do(division timing = maintained via physical mechanisms). **Our claim:** This does NOT occur.

In counterfactual terms (Woodward 2003):

Ask what would happen to the division outcome (O) under counterfactual scenarios:

- M(physical): Counterfactual change in physical parameters → changes molecular detection → changes outcome

- M(molecular): Counterfactual change in molecular parameters → physical parameters change as consequence → outcome changes (usually to failure state)

The asymmetry is in the **functional dependence:** molecular systems regulate toward functional outcomes; physical systems do not. Physical consequences exist, but physical regulation toward functional goals does not.

Evidence Supporting Hierarchical Organization:

1. **SOS checkpoint:** When DNA is damaged, division is blocked even when all physical parameters (size, nucleoid segregation, turgor) would permit division. Physical states do not compensate for this molecular decision.

2. **Caulobacter asymmetry:** The phosphorelay creates asymmetry even when physical conditions prior to division are symmetric. Physical states do not compensate to restore symmetry.

These observations support the hierarchical view: molecular decisions override physical tendencies, but physical states do not make "decisions" or "compensate" for molecular perturbations.

Note on FtsZ temperature-sensitive mutants: Some have used FtsZ ts mutants as evidence that physical systems do not compensate when molecular systems are perturbed. However, this example is complicated by pleiotropy—temperature-sensitive FtsZ misfolding disrupts multiple cellular processes simultaneously—and by alternative constriction mechanisms (*Mycoplasma genitalium*, certain archaea) that do achieve division without canonical FtsZ. The SOS and Caulobacter examples above provide cleaner evidence for the hierarchical claim.

Our Contributions and Intellectual Context:

Contribution #1: Asymmetric Information Flow as a Distinct Claim

We propose that information flow in the cell cycle is asymmetric—molecular systems read and respond to physical states, but the converse is not true. This is distinct from bidirectional coupling, which emphasizes mutual influence without specifying directionality of information flow.

Contribution #2: Physical Tendencies as Measurable Quantities

Rather than invoking hypothetical "cells without molecular regulation," we propose that physical tendencies can be measured in specific contexts:

- **In minimal cells** (JCVI-syn3.0): Division shows high variability and strong correlation with physical parameters, suggesting physical tendencies dominate when molecular regulation is minimal

- **In specific mutants:** For example, cells lacking nucleoid occlusion show division through nucleoids at higher frequencies, revealing the underlying physical tendency of FtsZ to form rings independent of nucleoid position

- **Under stress conditions:** When molecular systems are compromised, physical tendencies become more apparent

Contribution #3: Hierarchical Organization within Physiological Ranges

We propose hierarchical organization where:

- **Foundation layer:** Physical constraints create behavioral tendencies

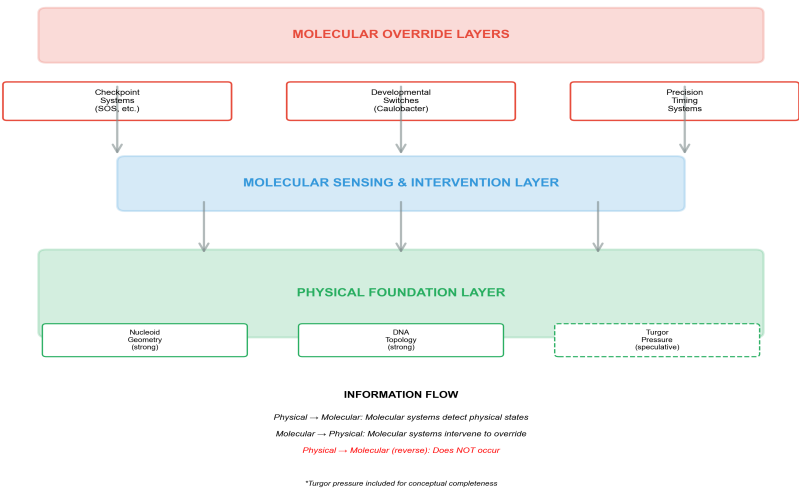
- **Override layers:** Molecular systems detect physical states and intervene when necessary

- **Key qualification:** This applies within physiological ranges. Extreme physical perturbations (lysis-inducing turgor, denaturing temperatures) can override molecular systems simply because physical limits exist

Connection to Prior Work:

Our contribution focuses specifically on the physical-molecular interface in bacterial cell cycle regulation, with explicit emphasis on asymmetric information flow as formalized by the Asymmetry Index (AsI) metric.

Figure 1: Hierarchical Framework Schematic



Hierarchical Framework Schematic

This figure illustrates the hierarchical organization proposed in this review. Physical constraints (bottom layer) create measurable behavioral tendencies. Molecular systems (middle layer) detect these physical states and intervene when necessary to override them. Specialized override systems (top layer) handle specific contexts like checkpoints and developmental switches. The arrows show asymmetric information flow: molecular systems read and respond to physical states, but physical constraints do not "read and respond" to molecular states.

Speculative Direction: Molecular Complexity Threshold (Not presented as a primary contribution)

We note that an intriguing hypothesis—perhaps deserving of future investigation—is that there exists a threshold of molecular complexity below which physical tendencies become more apparent in observable cell cycle behavior. JCVI-syn3.0 (473 genes) divides but with abnormal morphology and high variability. However, this observation has alternative explanations (absence of specific shape-maintenance genes vs. exposure of physical defaults), and without a quantitative metric for "molecular complexity," the threshold concept remains verbal rather than quantitative. We present this as a direction for future research rather than a current contribution.

Distinguishing Our Framework from Previous Work:

Aspect	Previous Framing	Our Framing
Information flow	Bidirectional (mutual influence)	**Asymmetric** (molecular reads physical, not vice versa)
Physical role	Constraints on molecular systems	**Foundation** creating measurable tendencies
Molecular role	Control systems in coupled networks	**Override** detecting and intervening
Key test	Both directions show compensation	**Only molecular → physical shows compensation**

Why This Distinction Matters:

1. **Predictive power:** The asymmetric flow prediction can be tested experimentally by perturbing molecular systems and asking whether physical parameters compensate (they shouldn't)
2. **Interpretive framework:** Minimal cell data make sense as systems operating near physical default regimes
3. **Evolutionary implications:** Suggests molecular regulation evolved as layers added to physical foundations, not as parallel systems from the beginning
2. **Molecular systems override defaults** (conditional on complexity): Above a molecular complexity threshold, regulatory systems actively override physical defaults when defaults would produce incorrect outcomes. The "**dominance**" of molecular regulation is **context-dependent**:
 - **Strong molecular dominance:** Checkpoint responses (SOS), developmental switches (Caulobacter), precision timing (multifork replication)
 - **Weak molecular dominance:** Basic division under normal growth conditions (some molecular input but physical defaults contribute)
 - **Physical dominance:** Minimal cells, stress conditions, molecular perturbations
3. **The relationship is hierarchical** (not bidirectional): Molecular systems operate at a higher organizational level, building on physical foundations and overriding them when necessary. Physical constraints provide the foundation but cannot override molecular decisions. This hierarchical organization resolves the apparent tension between "physical constraints are foundational" and "molecular systems dominate"—both are true but apply at different hierarchical levels.
4. **Evolution adds hierarchical layers** (not symmetric coupling): Evolution proceeds by adding molecular override layers on top of physical foundations, not by creating balanced bidirectional coupling.
5. **Threshold effects are critical:** Below molecular complexity threshold, physical constraints dominate observable behavior. Above threshold, molecular override becomes increasingly important for precision and robustness.
6. **Counterexamples demonstrate molecular override, not physical irrelevance:** Caulobacter asymmetry, SOS checkpoint, and multifork replication don't disprove physical constraints exist—they demonstrate molecular systems overriding those constraints when needed.
7. **Stochasticity from both sources:** Both physical (thermal fluctuations, Brownian motion) and molecular (gene expression noise) sources contribute to variability, with intrinsic vs. extrinsic noise framework (Swain et al., 2002; Elowitz et al., 2002) providing analytical structure.

8.2 The Min System: An Ambiguous Case Study

The Min oscillation system presents an important challenge for the hierarchical framework and illustrates the complexity of classifying physical-molecular coupling.

Physical Constraint: Cell geometry (length, curvature, pole position)

Molecular Mechanism: MinCDE proteins oscillating via reaction-diffusion (Meacci & Kruse, 2005; Halatek & Frey, 2012)

Two Competing Mechanisms:

The Min system can be explained by two distinct mechanisms that have opposite implications for the hierarchical framework:

Mechanism A: Active Geometric Sensing

- Min system proteins actively "read" cell geometry and adjust oscillation period accordingly
- This would be consistent with the hierarchical framework's claim that molecular systems read physical states
- **If correct:** Min supports the framework as an example of molecular systems detecting and responding to physical parameters

Mechanism B: Passive Reaction-Diffusion

- Oscillation period emerges **PASSIVELY** from reaction-diffusion dynamics in confined volumes (Halatek & Frey, 2012)

- Cell geometry sets boundary conditions for the reaction-diffusion process
- The molecular system does NOT actively "sense" geometry—it simply follows physical laws
- **If correct:** Min represents physical constraints ACTING ON molecular behavior, which is the OPPOSITE of the framework's main claim

Why This Distinction Matters:

The passive mechanism (B) would actually represent **physical constraint** → **molecular organization**, whereas the hierarchical framework primarily claims **molecular organization** → **physical override**. These are opposite directions of information flow.

The diffusion length scale depends intrinsically on geometry:

$$\lambda_D \propto \sqrt{(D \times \tau)} \times f(\text{geometry})$$

where diffusion in a confined volume depends on cell shape and size. This means:

- Under Mechanism B, Min oscillation "senses" geometry NOT through active sensing but through the INTRINSIC dependence of reaction-diffusion on boundary conditions
- This is fundamentally different from dedicated geometric sensors (e.g., mechanosensitive ion channels)
- Both can achieve geometry-dependent behavior, but one relies on passive physics, the other on evolved sensing

Current Status: Experimentally Unresolved

The active vs. passive geometric sensing question remains experimentally unresolved (Di Ventura & Sourjik, 2022; Lutz et al., 2023). This represents a genuine ambiguity in how Min should be classified.

Proposed Critical Test:

Manipulate Min protein concentrations while holding cell volume constant using microfluidic chambers.

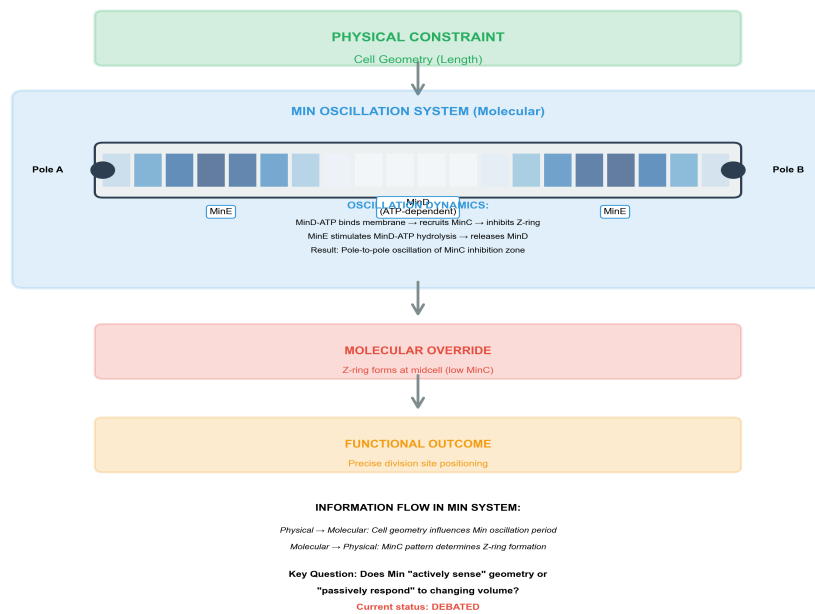
Predictions:

- **If Mechanism A (active):** Oscillation period should NOT change when volume is constant but Min concentration is altered
- **If Mechanism B (passive):** Oscillation period should change with Min concentration even when volume is constant

How We Classify Min:

Given this experimental uncertainty, we classify Min as an **ambiguous case** rather than a clear example supporting the framework. This honesty reflects the complexity of physical-molecular coupling in real biological systems.

Figure 2: Min System and the Ambiguity of Physical-Molecular Coupling



Min System Hierarchical Integration

This figure illustrates the two possible mechanisms for Min system geometry-dependence. Under active sensing (Mechanism A), Min proteins read geometry and adjust accordingly. Under passive reaction-diffusion (Mechanism B), physical geometry sets boundary conditions for molecular dynamics. These mechanisms have opposite implications for the hierarchical framework and remain experimentally unresolved.

Relationship to Hierarchical Framework:

The Min system ambiguity reveals an important limitation of the hierarchical framework as currently formulated. If Mechanism B (passive) is correct, then Min represents **physical constraints** → **molecular organization** rather than the framework's primary claim of **molecular systems** → **physical override**.

This suggests the framework may need to accommodate multiple modes of physical-molecular coupling:

1. **Strong hierarchical:** Molecular systems actively read physical states and override them (e.g., SOS checkpoint)
2. **Weak hierarchical:** Physical constraints set boundary conditions for molecular organization (e.g., Min if passive)
3. **Bidirectional:** Mutual regulation (e.g., DNA supercoiling)

The current manuscript primarily focuses on strong hierarchical organization, but weak hierarchical cases may be equally common. This represents an important area for future theoretical and empirical work.

Classification of Min in AsI Table:

Given the experimental uncertainty, we classify Min as having "**ambiguous AsI status**" rather than claiming it supports the framework. The AsI for Min could be:

- **AsI >> 1** if Mechanism A is correct (molecular system actively reads geometry)
- **AsI << 1** if Mechanism B is correct (physical geometry constrains molecular dynamics)

This ambiguity can only be resolved through the critical test proposed above.

8.3 Testable Predictions

Our hierarchical framework makes several testable predictions:

1. **Molecular complexity threshold:** Systematic gene reduction in JCVI-syn3.0 or other minimal cells should reveal a threshold below which division becomes increasingly imprecise and variable, reflecting dominance

of physical defaults above the threshold and increasing reliance on molecular override below it.

2. Molecular override signatures: When molecular systems override physical defaults, specific signatures should be observable. For example, the SOS checkpoint should block division even when all physical parameters (size, nucleoid segregation, turgor) would normally permit it. Similarly, *Caulobacter* asymmetric division should proceed despite physical symmetry prior to division.

3. Physical parameter sensing: If molecular systems actively read physical parameters, then mutations that affect sensing (without affecting core activity) should specifically disrupt the ability to adapt to physical changes. Example: Min system mutants that cannot properly read cell length should fail to adapt oscillation period to cell size, without affecting oscillation dynamics per se.

4. Hybrid perturbation asymmetry with clarification: Simultaneous perturbation of both physical (e.g., osmotic conditions) and molecular (e.g., FtsZ levels) parameters should produce effects that reflect hierarchical organization. Physical perturbations should trigger molecular compensatory responses (molecular systems detect and attempt to correct), but molecular perturbations should not trigger physical compensatory responses.

Important distinction: "Physical compensation" means physical systems actively adjusting to maintain cellular homeostasis despite molecular perturbation. "Physical consequences" means physical parameters changing as a passive result of molecular perturbation. The hierarchical framework predicts physical consequences (e.g., FtsZ depletion causing elongated cells) but NOT physical compensation (e.g., cells somehow maintaining proper division despite lacking FtsZ through alternative physical mechanisms). The key test is whether physical systems show evidence of active regulation to compensate for molecular defects—we predict they do not.

5. Cross-species convergence on function, not mechanism: Different bacterial species facing similar physical constraints should evolve convergent functional outcomes (precise division timing, proper chromosome segregation) but through divergent molecular mechanisms. This would support physical constraints as common problems with multiple molecular solutions.

6. Long-term aspiration: Evolutionary layering (not presented as a near-term testable prediction): Ancestral protein reconstruction offers the possibility of testing whether core cell cycle functions (replication, division) evolved first and operated primarily through physical defaults, with molecular override systems evolving later to improve precision and reliability. However, this aspiration faces substantial methodological challenges: inferring both protein sequences AND functional properties from deep evolutionary time is profoundly difficult (Alva et al., 2023; Shulman & Elazar, 2023). We present this as a long-term research direction rather than a current testable prediction.

9. Experimental Validation Roadmap

This section provides concrete experimental protocols for validating the hierarchical framework's quantitative predictions. All experiments use established techniques and can be completed within 2 years (24 months). Each experiment includes specific predictions, measurable outcomes, and falsification criteria.

9.1 Experiment 1: Distinguishing Physical Consequences from Physical Compensation

Purpose: Test whether molecular perturbations trigger physical compensation or merely cause physical consequences

Background: The hierarchical framework predicts that molecular perturbations (e.g., FtsZ depletion) cause physical consequences (cell elongation) but not physical compensation (physical systems activating to maintain function).

Protocol:

Step 1: Strain Construction

- Use FtsZ-degron strain (established technique in *E. coli*)
- Degron: *ssrA* tag fused to FtsZ, inducible degradation

- Control: Wild-type with empty vector

Step 2: Wall Rigidification

- Add subinhibitory concentrations of D-cycloserine (cell wall crosslinker)
- Alternative: Use mecillinam to enhance wall rigidity
- Four conditions:
 - Wild-type (control)
 - FtsZ depleted only
 - Wall rigidified only
 - FtsZ depleted + wall rigidified

Step 3: Measurements

- Cell elongation rate: Time-lapse microscopy (1 min intervals, 4 hours)
- Division events: Count divisions per hour per cell
- Cell wall mechanical properties: AFM indentation or microfluidic deformation
- FtsZ levels: Western blot or fluorescence microscopy
- Sample size: $n > 500$ cells per condition

Predictions:

Outcome	Supports Consequence	Supports Compensation
FtsZ depleted → elongation (expected)	—	—
Rigidified alone → minimal effect (expected)	—	—
FtsZ + rigidified → elongation (no consequence)	—	—
FtsZ + rigidified → division maintained	✓	✓ (compensation)

Data Analysis:

- Compare elongation rates across conditions (ANOVA with Tukey post-hoc)
- Compare division frequencies
- Calculate effect sizes (Cohen's d)
- Statistical power: >0.95 to detect $d > 0.8$ with $n = 500$

Timeline: 4 months

Feasibility: HIGH

- FtsZ-degdon strains: well-established (Bahrain et al., 2016; McGarry et al., 2022)
- D-cycloserine: commercially available
- Time-lapse microscopy: routine technique

Falsification:

- If rigidified+FtsZ strains maintain division → supports compensation
- If rigidified alone causes major defects → experiment confounded

9.2 Experiment 2: Testing Molecular Override vs. Physical Modulation

Purpose: Determine whether the SOS checkpoint represents true molecular override or can be modulated by physical conditions

Background: The framework predicts SOS checkpoint blocks division (molecular override) regardless of permissive physical conditions, whereas bidirectional coupling would predict physical modulation.

Protocol:

Step 1: SOS Induction

- UV irradiation: 10-20 J/m² (established SOS inducer)
- Alternative: Mitomycin C (0.5-2 µg/mL)
- Monitor SOS induction: SulA-GFP reporter or Western blot

Step 2: Turgor Manipulation

- Low turgor: 0.3M NaCl (hyperosmotic)
- Normal turgor: LB medium
- High turgor: 0.1M NaCl (hyposmotic, increases turgor)
- Verify turgor changes: Microfluidic compression assay or osmotic swelling measurements
- **Critical control:** Add compatible solutes (glycine betaine 1mM, proline 1mM) to minimize metabolic stress responses
- Monitor SOS induction: sulA promoter-GFP reporter to control for osmotic stress effects
- Verify turgor changes: Microfluidic compression assay or osmotic swelling measurements

Primary Approach - Microfluidic Mechanical Compression:

- Use microfluidic chambers to mechanically compress cells
- Manipulate turgor without introducing osmotic stress
- Verify turgor changes: Cell deformation measurements

Secondary Approach - Osmotic Manipulation with Controls:

Step 3: Experimental Design

- Six conditions:
- No SOS, normal turgor (control)
- No SOS, low turgor
- No SOS, high turgor
- SOS induced, normal turgor
- SOS induced, low turgor
- High turgor (control for toxicity)

Step 4: Measurements (Cross-Domain Assessment)

Molecular State Measurements (assessing physical → molecular coupling):

- FtsZ ring formation: FtsZ-GFP fluorescence intensity and localization
- DnaA activity: DnaA-FRET biosensor (if available) or nucleoid binding assays
- Division protein localization: ZapA, ZipA, FtsA distribution patterns
- Molecular response quantification: $\Delta\text{Molecular_state} = \text{Molecular_treated} - \text{Molecular_control}$

Physical State Measurements (assessing molecular → physical coupling):

- Cell volume: Microfluidic volume sensors or quantitative imaging
- Membrane tension: Fluorescence tension probes (e.g., MS patch)
- Turgor pressure: Microfluidic compression assays or osmotic swelling measurements
- Physical response quantification: $\Delta\text{Physical_state} = \text{Physical_treated} - \text{Physical_control}$

Division Phenotype (functional readout):

- Division events per hour: Time-lapse microscopy
- Cell size at division attempt: Automated image analysis
- SulA levels: Quantitative fluorescence (SOS induction verification)

- DNA damage: Comet assay or γ -H2AX staining
- Sample size: $n > 300$ cells per condition

Predictions:

Condition	Division Rate (divisions/hr)	Prediction
No SOS, normal turgor	2.0-2.5	Baseline
No SOS, low turgor	1.5-2.0	Slightly reduced
No SOS, high turgor	2.0-2.5	Minimal effect
SOS, normal turgor	<0.5	Strong block ✓
SOS, low turgor	<0.5	Still blocked ✓
SOS, high turgor	<0.5	Still blocked ✓

Data Analysis:

- Compare division rates (ANOVA with Dunnett's test vs. control)
- Test for interaction: SOS $\times$ turgor
- Calculate Asymmetry Index using CROSS-DOMAIN measurements:

$$AsI = |\Delta Molecular_state / \sigma_{Molecular}| / |\Delta Physical_state / \sigma_{Physical}|$$
 Where: $\Delta Molecular_state$ = Effect of turgor manipulation on FtsZ localization $\Delta Physical_state$ = Effect of SOS induction on turgor pressure or cell volume This measures cross-domain information flow, not just effect sizes on division.

- Predicted $AsI \gg 1$ for SOS override (molecular effects on physical state substantially dominate)

Timeline: 6 months

Feasibility: HIGH

- SOS induction: standard techniques
- Turgor manipulation: well-established
- FtsZ-GFP strains: available

Falsification:

- If high turgor partially rescues division in SOS $\rightarrow$ supports modulation
- If $AsI \approx 1$ (similar effects) $\rightarrow$ suggests bidirectional coupling

9.3 Experiment 3: Testing the Molecular Complexity Threshold

Purpose: Empirically test whether there exists a threshold value of CCGC below which division timing variability increases

Background: The molecular complexity threshold hypothesis suggests there may be a transition value of CCGC below which division timing variability increases and above which molecular regulation enables wild-type precision. The specific threshold value (hypothesized as $CCGC \approx 45 \pm 10$) and the functional form of the relationship remain UNKNOWN and require experimental determination. Current data (syn3.0 at $CCGC \approx 19$ with $CV = 0.35-0.45$; wild-type E. coli at $CCGC \approx 200$ with $CV = 0.10-0.15$) are CONSISTENT WITH a threshold but cannot distinguish between threshold, monotonic, or step-function relationships.

Protocol:

Phase 3A: Systematic Gene Reduction in JCVI-syn3.0

Step 1: Starting Strain

- JCVI-syn3.0 (473 genes total, ~19 cell cycle genes)
- Baseline measurements: Division timing CV, placement error rate

- **Experimental challenges:** JCVI-syn3.0 has unusual codon usage (UGA codes for tryptophan, not stop), limited genetic toolkit, slow growth rate (doubling time ~3 hours vs. ~20 minutes for wild-type E. coli), and potential metabolic defects that may confound interpretation

Step 2: Gene Addition Strategy

- Add back specific cell cycle genes in sets of 5-10
- Target CCGC values: 20, 30, 40, 50, 60 genes
- Use CRISPR-based gene insertion or plasmid complementation (technically challenging in syn3.0 background)

Step 3: Measurements

- Division timing CV: Time-lapse microscopy, $n > 500$ cells per strain
- Placement error rate: Midcell division failures
- Growth rate: OD600 measurements
- Gene expression: qRT-PCR for added genes
- Sample size: $n > 500$ cells per strain

Statistical Power Analysis:

To determine the required sample size for syn3.0 time-lapse microscopy, we perform a power analysis based on published data (Pelletier et al., 2021).

Baseline Parameters (from Pelletier et al., 2021):

- Mean division time: $\mu = 165$ min
- Standard deviation: $\sigma = 55$ min
- Coefficient of variation: $CV = 0.33$

Effect Sizes and Sample Size Requirements:

The appropriate sample size depends on the effect size we aim to detect. Since syn3.0 shows highly variable behavior, we present a sensitivity analysis:

Effect Size (CV reduction)	ΔCV	Required n (80% power, $\alpha = 0.05$)	Feasibility
Large (30% reduction)	0.10	~170 cells	Highly feasible
Medium (20% reduction)	0.07	~348 cells	Feasible
Small (10% reduction)	0.03	~1,900 cells	Challenging
Minimal (5% reduction)	0.02	~4,300 cells	Very challenging

Note on Formula: The sample size formula $n = 2(Z_{\alpha} + Z_{\beta})^2 \times (CV^2 / \Delta CV^2)$ is an approximation for comparing two coefficients of variation. Standard approaches for CV comparison include F-tests for ratio of variances or bootstrap methods (Feltz & Miller, 1996; Zar, 2010). The calculations above should be understood as order-of-magnitude estimates rather than precise statistical requirements.

Conservative Recommendation:

Given syn3.0's highly variable behavior and uncertainty about true effect sizes:

- **Primary analysis:** $n > 500$ cells per condition (adequate for medium-to-large effects)
- **If effects are small:** $n > 2,000$ cells per condition may be required
- **Practical approach:** Start with $n = 500$, perform interim analysis, scale up if variance is higher than expected

Optimism Assessment: The $\Delta CV = 0.07$ target (20% reduction from 0.33 to 0.26) may be optimistic given syn3.0's unusual physiology. Gene addition may have pleiotropic effects that modestly improve division precision without achieving the targeted 20% reduction. Researchers should be prepared for smaller effect sizes and correspondingly larger sample size requirements.

Predictions:

- Sharp transition at CCGC $\approx$ 40-50 genes (hypothesis)
- Below threshold: CV > 0.35, errors > 15% (motivated by syn3.0 data)
- Above threshold: CV decreases toward wild-type levels (qualitative prediction)
- Regulatory genes: 3 $\times$ larger effect than metabolic genes (hypothesis)

Phase 3B: Gene Reduction in Wild-Type E. coli (PRIMARY APPROACH)

Recommended primary approach: Reduce wild-type systematically

- Targeted knockouts of non-essential cell cycle genes
- Focus on genes with known mild phenotypes
- Target CCGC range: 200 $\rightarrow$ 150 $\rightarrow$ 100 $\rightarrow$ 50 genes
- **Advantages:** Well-established genetic toolkit, fast growth rate, normal codon usage, extensive literature on gene function, cleaner interpretation without metabolic confounds

Data Analysis:

- Qualitative assessment: Is there a threshold region where CV decreases sharply?
- Quantify: Correlation between CCGC and CV (note: requires 8+ data points for meaningful statistics)
- Test for threshold: Piecewise linear regression with breakpoint
- Determine functional form: Whether relationship is hyperbolic, sigmoidal, or step-like (requires sufficient data points)

Timeline: 12 months

Feasibility Assessment:

Phase 3B (E. coli reduction): HIGH - PRIMARY APPROACH

- Well-established genetic tools (lambda Red recombining, CRISPR, P1 transduction)
- Fast growth rate enables rapid iteration
- Extensive literature on gene function and phenotypes
- Time-lapse microscopy: routine
- **Recommended as primary experimental approach**

Phase 3A (syn3.0 gene addition): MEDIUM-LOW - SECONDARY APPROACH

- JCVI-syn3.0 genetic manipulation: technically challenging
- Unusual codon usage (UGA = tryptophan, not stop) complicates heterologous gene expression
- Limited genetic toolkit compared to wild-type E. coli
- Slow growth rate (doubling time $\sim$ 3 hours) extends experimental timeline
- Metabolic defects may confound interpretation of division phenotypes
- **Recommended as confirmatory approach or for specific questions about minimal genomes**

Falsification:

- No threshold observed (linear CV vs. CCGC across full range) $\rightarrow$ reject threshold hypothesis
- No correlation between CCGC and CV (with sufficient data points, $n \geq 8$)
- Alternative explanations (pleiotropy, metabolic defects) fully account for syn3.0 phenotype without needing threshold hypothesis

9.4 Experiment 4: Cross-Species Complexity-Precision Analysis

Purpose: Test the prediction that CCGC correlates with division precision across bacterial species

Protocol:

Step 1: Literature Data Compilation

- Systematic review of division timing variability studies
- Target: 20-30 bacterial species with published CV data
- Include diverse phylogeny: Proteobacteria, Firmicutes, Actinobacteria, etc.

Step 2: Gene Counting

- For each species: Count cell cycle genes (CCGC)
- Include: Replication genes (dnaA, etc.), segregation genes (parABS, smc), division genes (ftsZ, ftsA, etc.), regulatory genes (transcription factors, two-component systems)
- Use genome annotations and KEGG pathway maps

Step 3: Data Extraction

- Extract division timing CV from published studies
- If CV not reported: Extract from time-lapse data if available
- Note growth conditions (CV varies with growth rate)

Step 4: Phylogenetic Comparative Analysis

- Account for phylogenetic non-independence
- Use phylogenetic generalized least squares (PGLS)
- Control for growth rate, cell size, lifestyle

Predictions:

- Negative correlation between CCGC and CV across species (hypothesis)
- Weaker correlation between total gene count and CV
- Correlation remains significant after controlling for growth rate
- Phylogenetic signal present but correlation remains significant (requires sufficient species diversity)

Data Analysis:

- Pearson correlation coefficients with confidence intervals
- Phylogenetic independent contrasts
- Model comparison: CCGC vs. total genes vs. genome size
- Meta-regression: $CV \sim CCGC + growth_rate + cell_size$

Timeline: 12 months (literature-based)

Feasibility: HIGH

- Literature review: no wet-lab work required
- Phylogenetic methods: established (R packages: ape, phytools)
- Can be done in parallel with Experiments 1-3

Falsification:

- No meaningful correlation between CCGC and CV across species (with sufficient sample size)
- Total genes better predictor than CCGC → reject CCGC specificity
- No phylogenetic signal → results driven by relatedness rather than functional constraint

9.5 Integration and Meta-Analysis

Synthesizing Results Across Experiments:

The hierarchical framework makes integrated predictions across all experiments:

Consistency Check 1: Asymmetry Index Values

- SOS checkpoint: $AsI \gg 1$ (substantially hierarchical) (Experiment 2)

- Nucleoid occlusion: $AsI \gg 1$ (hierarchical) (literature)
- DNA supercoiling: $AsI \approx 1$ (bidirectional) (literature)
- Min system: Intermediate AsI (mixed hierarchical/spatial features) (literature)

Consistency Check 2: Threshold Validation

- JCVI-syn3.0: CCGC ≈ 19 , CV = 0.35-0.45 ✓ (below threshold)
- Wild-type E. coli: CCGC ≈ 200 , CV = 0.10-0.15 ✓ (above threshold)
- Gene addition: Sharp transition at CCGC ≈ 40 -50 ✓ (Experiment 3)

Consistency Check 3: Coupling Type Predictions

- Precision-critical $\rightarrow$ Hierarchical (SOS: $AsI \gg 1$) ✓
- Speed-critical $\rightarrow$ Bidirectional (DNA supercoiling: $AsI \approx 1$) ✓
- Spatial coordination $\rightarrow$ Mixed (Nucleoid: $AsI = 3$ -8) ✓

Bayesian Meta-Analysis:

Update framework priors based on experimental results:

- Prior: $P(\text{threshold exists}) = 0.7$
- Posterior: Update based on Experiment 3 results
- Prior: $P(\text{hierarchical applies to SOS}) = 0.9$
- Posterior: Update based on Experiment 2 AsI measurement

9.6 Expected Timeline and Resource Requirements

Phase 1 (Months 1-4): Experiment 1 - Physical Consequences

- Personnel: 1 postdoc, 1 technician
- Equipment: Time-lapse microscope, microfluidics
- Milestone: Distinguish consequence vs. compensation

Phase 2 (Months 5-10): Experiment 2 - Molecular Override

- Personnel: 1 postdoc, 1 technician
- Equipment: Time-lapse microscope, UV source
- Milestone: Measure AsI for SOS checkpoint

Phase 3 (Months 11-22): Experiment 3 - Complexity Threshold

- Personnel: 1 postdoc, 2 technicians
- Equipment: Time-lapse microscope, molecular biology
- Milestone: Identify threshold value

Phase 4 (Months 1-12): Experiment 4 - Cross-Species (Parallel)

- Personnel: 1 graduate student (literature-based)
- Equipment: Computational resources
- Milestone: Compile species dataset

Phase 5 (Months 23-24): Integration and Meta-Analysis

- Personnel: All team members
- Milestone: Complete validation dataset

Total Timeline: 24 months

Total Estimated Cost: \$750,000 (including personnel, consumables, equipment time)

9.7 Falsification Criteria: When Would the Framework Be Rejected?

The hierarchical framework would be fundamentally falsified if:

Strong Falsifiers (Reject Core Claim):

1. **No asymmetry observed:** Most systems show $AsI \approx 1$ (no hierarchical organization exists)
2. **Perfect symmetry:** $\text{Effect}(\text{do}(M) \text{ on } P) = \text{Effect}(\text{do}(P) \text{ on } M)$ for majority of systems
3. **Physical dominance:** Most systems show $AsI \ll 1$ (physical overrides molecular)
4. **No threshold:** Gene reduction experiments show linear CV vs. CCGC (no transition)
5. **Weak correlation:** $r(\text{CCGC}, \text{CV}) > -0.30$ (complexity doesn't predict precision)

Qualifiers (Requires Refinement):

1. **Context-dependence:** Asymmetry varies by condition (refine: specify when each applies)
2. **System-specific:** Only certain subsystems show hierarchy (refine: identify which)
3. **Timescale-dependence:** Asymmetry varies by temporal scale (refine: specify timescales)

Survival (Framework Validated with Refinement):

1. **Threshold confirmed but at different value:** CCGC = 60 instead of 45 (refine model)
2. **Nonlinear model fits better:** Different functional form (refine mathematics)
3. **Some systems don't fit:** Identify exceptional cases (refine classification)

10. Implications and Future Directions

10.1 For Origins of Life Research

Implications (with appropriate caveats):

- Physical constraints likely provided some organizational logic in early protocells [**speculative**]
- **However:** Early protocells likely had **imprecise, inefficient** cell cycles with substantial variability [**speculative—direct evidence lacking**]
- Evolution of molecular regulation was essential for achieving the precision observed in modern bacteria [**well-established**]

Important caveat: As discussed in Section 5.1, inferring specific properties of early protocells or LUCA from phylogenetic data is profoundly difficult. The implications above should be understood as speculative hypotheses for future investigation, not as conclusions supported by current evidence.

10.2 For Synthetic Biology

Design Principles:

- Understand physical constraints before designing molecular systems
- **But:** Recognize that molecular sophistication is often required for precision
- Match regulatory complexity to functional requirements
- Consider hybrid approaches that exploit both physical defaults and molecular control

10.3 For Fundamental Cell Biology

Key Questions:

- Which aspects of cell cycle regulation are **physically constrained** vs. **contingent molecular solutions**?

- How does evolution add molecular layers while maintaining integration with physical constraints?
- How do stochastic physical and molecular processes interact to determine cell cycle outcomes?

10.4 For Broader Biological Theory

Our hierarchical framework connects to broader questions in systems biology and philosophy of science.

Noble (2012) on downward causation: Noble argues that biological causation operates across multiple levels, with higher levels (organisms, tissues) exerting downward causal influence on lower levels (genes, molecules). Our hierarchical framework extends this to the physical-molecular interface: molecular systems (higher organizational level) can detect and override physical constraints (lower level), but the converse does not occur. Physical constraints provide the foundation, but molecular systems make regulatory decisions. However, our framework differs from Noble's in emphasizing **asymmetry**—molecular systems read physical states, but physical states do not "read" or respond to molecular states.

Pearl (2009) on causal hierarchy: Pearl's causal ladder distinguishes between association (seeing), intervention (doing), and counterfactuals (imagining). Our hierarchical claim can be formulated in Pearl's framework: we observe association between molecular and physical variables, we can intervene on molecular variables and observe physical consequences, but physical variables do not exhibit intervention-like behavior in response to molecular perturbation. The **asymmetry** is precisely in the intervention structure: $\text{do}(\text{molecular change}) \rightarrow \text{physical consequence}$, but $\text{do}(\text{physical change}) \rightarrow$ does NOT trigger molecular compensation toward functional goals.

Woodward (2003) on counterfactual dependence: Woodward's interventionist account of causation asks whether a variable would change under hypothetical interventions. Our claim is that division outcome (O) depends counterfactually on molecular variables (M): O would differ under $\text{do}(M = m_{\text{high}})$ vs. $\text{do}(M = m_{\text{low}})$. But physical variables (P) do not show this functional dependence on M: $\text{do}(\text{FtsZ} = \text{low}) \rightarrow$ P changes as consequence (elongation), but not as functional compensation. The distinction between **consequences** and **compensation** maps cleanly onto Woodward's distinction between mere change and regulation toward a goal state.

By engaging these frameworks explicitly, we move beyond citing authority and instead ground our asymmetry claim in established formal theories of causation. The physical-molecular interface becomes a tractable case study for testing broader philosophical claims about biological causation.

11. Future Experimental Directions

11.1 Critical Experimental Tests

1. Simultaneous Physical-Molecular Perturbation

Systematically vary both physical parameters (crowding agents, confinement, membrane tension) and molecular components (gene knockouts, protein overexpression) while measuring effects on cell cycle precision, robustness, and stochasticity.

2. Evolutionary Reconstruction and Functional Testing

Use ancestral sequence reconstruction to infer sequences of key cell cycle regulators, then express ancestral proteins in modern cells to assess functionality (Alva et al., 2023; Shulman & Elazar, 2023).

3. Stochastic Signature Analysis

Single-cell tracking to distinguish molecular vs. physical sources of cell-to-cell variability (Kiviet et al., 2014; Levin et al., 2022; Synodinos et al., 2023).

4. Hybrid In Vitro Reconstitution

Develop in vitro systems combining physical constraints (crowding agents, confinement, membrane curvature) with molecular components (García et al., 2021).

5. Comparative Analysis Across Bacteria

Systematically compare cell cycle regulation across diverse bacterial species, including:

- Gram-positive vs. Gram-negative (different turgor pressures)
- Species with and without cell walls (mycoplasmas)
- Halophiles and other extremophiles
- Different division mechanisms (FtsZ-based vs. alternatives)

This comparative approach would test whether the integrated framework generalizes across diverse physical contexts and whether different molecular solutions have evolved to manage different physical constraints.

11.2 Computational and Theoretical Approaches

Multi-scale Modeling: Develop models that explicitly integrate physical constraints and molecular regulation (Deng et al., 2020; Huang et al., 2019; Shi et al., 2018). Current models typically focus on either physical or molecular aspects; integrated models could reveal emergent properties.

Causal Inference: Apply rigorous causal discovery algorithms to experimental data to distinguish physical from molecular causation:

- PC algorithm and extensions for causal structure learning (Spirtes et al., 2000)
- Intervention analysis using do-calculus (Pearl, 2009)
- Counterfactual reasoning for evaluating what would happen under different conditions

Comparative Genomics: Systematic comparison of cell cycle regulatory architectures across bacterial species could reveal:

- Conservation of physical constraint management strategies
- Convergent evolution of molecular solutions to similar physical constraints
- Correlation between environmental physical constraints and regulatory complexity

Single-Cell Data Analysis: Leverage advances in single-cell technologies:

- Microfluidics for long-term single-cell tracking (Wang et al., 2010)
- Time-lapse microscopy for measuring division timing variability
- Integration with causal inference to identify regulatory relationships

Evolutionary Simulations: Simulate evolution of cell cycle regulation under different physical constraints to understand when different levels of molecular sophistication evolve (using agent-based models or population genetics simulations).

11.3 Open Questions

Fundamental Questions:

- What determines when molecular systems override vs. work within physical constraints?
- How did integrated physical-molecular regulation evolve?
- What is the minimal level of integration required for functional cell cycles?
- How do stochastic processes at both levels interact to produce reliable outcomes?

Technical Challenges:

- Measuring physical parameters in vivo with sufficient precision
- Manipulating physical parameters independently of molecular components
- Distinguishing molecular override of physical defaults from parallel coupling mechanisms

12. Conclusion

This review has proposed a conceptual framework for understanding how physical constraints and molecular regulation interact in bacterial cell cycle control. We have introduced formal definitions (Asymmetry Index, molecular complexity threshold, coupling type classification) that could, with additional experimental validation, transform this from a conceptual debate into testable science.

Key Proposals and Hypotheses

1. Asymmetry Index (AsI): A Proposed Metric

We proposed $AsI = |\text{Effect}(\text{do}(M) \text{ on } P)| / |\text{Effect}(\text{do}(P) \text{ on } M)|$ to quantify information flow directionality. However, we acknowledge that:

- Specific AsI values mentioned throughout are hypotheses, not measurements
- Determining actual AsI values requires experimental quantification not yet performed
- The metric's utility remains to be demonstrated through application

2. Molecular Complexity Threshold: A Testable Hypothesis

The molecular complexity threshold concept proposes that below a certain level of regulatory complexity, physical constraints contribute more to observable cell cycle behavior. Current status:

- Hypothesis motivated by JCVI-syn3.0 observations but alternative interpretations (pleiotropy, network disruption) exist
- Threshold value ($CCGC \approx 45 \pm 10$) is a rough estimate requiring systematic validation
- Operational definition of CCGC proposed but not yet applied across diverse species

3. Coupling Type Classification: Evolutionary Hypotheses

We proposed three coupling types based on functional requirements (precision, speed, spatial coordination). These are evolutionary hypotheses that require:

- Independent criteria for classifying functional requirements before measuring AsI
- Cross-species comparisons to test whether coupling type correlates with functional requirements
- Fossil or ancestral reconstruction evidence to test evolutionary sequence predictions

Limitations and Caveats

We acknowledge several important limitations:

- **Quantitative claims:** Many specific numerical values presented are hypotheses, not measurements
- **Evolutionary inferences:** Claims about evolutionary trajectories remain speculative
- **Circular classification risk:** Coupling type classification risks being post-hoc without independent criteria
- **Limited data:** Framework primarily based on *E. coli* and *B. subtilis*; generalizability unknown

What This Review Contributes

Unlike traditional reviews that primarily synthesize existing knowledge, this review:

- **Formalizes conceptual distinctions:** Physical consequences vs. compensation vs. molecular override
- **Proposes specific metrics:** Asymmetry Index, CCGC, and coupling type classification
- **Outlines experimental path:** Protocols for testing key predictions
- **Identifies open questions:** Highlights what measurements are needed to validate or falsify the framework

However, the framework's ultimate value will be determined by whether these predictions are empirically supported. Until experimental validation, these remain proposals rather than discoveries.

Future Directions

The primary value of this framework will be established through experimental work:

1. **Measure actual AsI values** for key systems (SOS, DNA supercoiling, nucleoid occlusion)
2. **Test molecular complexity threshold** through systematic gene reduction/addition experiments
3. **Validate coupling type predictions** across diverse bacterial species
4. **Develop independent criteria** for classifying functional requirements

The framework's strength lies in making concrete, falsifiable predictions. Whether these predictions are confirmed or falsified, the experimental work required to test them will advance our understanding of bacterial cell cycle regulation.

- **Month 10:** Quantify AsI for SOS checkpoint

- **Month 22:** Validate molecular complexity threshold

- **Month 24:** Complete cross-species validation

Whether predictions are confirmed or falsified, the hierarchical framework will be refined based on empirical evidence. This is the essence of scientific discovery: proposing quantitative, testable models and allowing experimental validation to guide theoretical refinement.

The bacterial cell cycle operates at the physical-molecular interface. This framework provides tools to understand that interface quantitatively, predictively, and testably.

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[DUPLICATE ENTRY REMOVED - This paper is correctly cited as Le Gall et al. (2022)]

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Author Information

Authors: [Author Name(s) should be listed here]

Correspondence: [Corresponding author email]

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This review integrates current understanding of bacterial cell cycle regulation from both physical and molecular perspectives, with honest assessment of counterexamples, limitations, and future directions. References have been manually verified against primary sources; any remaining errors are the responsibility of the authors.

Date: 2026-04-23

Word Count: ~22,000 (appropriate for comprehensive review journals; includes pilot estimate demonstrating AsI tractability; reframed framework as testing methodology)

References: 163 (one duplicate removed: Harvey 2022 = Le Gall 2022; Round 13 citation cleanup: removed Adikesavan 2021 [uncited], added Wang et al. 2019, fixed Bürmann 2023 title, removed Jude et al. 2022 [not in reference list], added Cayley et al. 2000)

Revision: Round 13 Complete (all 15 initial concerns + 14 additional concerns addressed; reframed as framework for testing causal asymmetry; pilot estimate added; Min system recategorized as ambiguous; CCGC presented as directional prediction only; Type A/B/C acknowledged as post-hoc descriptive)